



one point 4 lines 12 neighbours 27 far points

Forty Points

One qutrit, the exceptional algebras, and a light cone

An exact and honest account of the $W(3,3)$ programme,
written so that anyone can follow it

The $W(3,3)$ Theory Programme

repository [wilcompute/W33-Theory](https://github.com/wilcompute/W33-Theory) · 25 September 2026

Every mathematical statement in this paper carries a status badge and, where it is a computation, the path of the committed script and certificate that proves it. Interpretations are labelled as interpretations. Things that were believed and later shown false are kept on the page, with the measurement that overturned them.

Abstract

For everyone. Take the smallest quantum system that has three settings instead of two — a *qutrit* — and let it act on itself. The ways it can be transformed, and the ways those transformations do or do not interfere with one another, form a finite geometric object with exactly forty points. This paper follows that single object as far as exact mathematics can take it. It turns out to contain a tiny version of space-time with a light cone; four interlocking “clocks”; the largest exceptional symmetry in mathematics, E_8 ; the octonionic algebra of 3×3 matrices that physicists have long suspected sits behind particle physics; and, inside that algebra, a genuine ten-dimensional Lorentz group whose spinors are the matter particles of one generation. We also say, just as clearly, what the object does *not* do: it does not by itself produce the masses, the forces, or the dynamics of our universe, and it provably cannot decide whether nature is left-handed or right-handed. Finally we settle, by an exhaustive exact computation, the most important open question about the string-theory models built on it: in their supersymmetric vacua the proton is not protected, because a field with the quantum numbers of a right-handed sneutrino is forced to condense.

For specialists. The symplectic generalized quadrangle $W(3,3)$ is the projective commutation geometry of the two-sided Weyl action on one qutrit’s Hilbert–Schmidt space; its Bell line is the unitary-channel diagonal. Fixing that line, the 27 transverse Lagrangians form $\text{Sym}_2(\mathbb{F}_3)$ with the finite Minkowski form $t^2 - x^2 - y^2$, whose null relation is context intersection; the four null directions are $\mathbb{P}^1(\mathbb{F}_3)$, the tetracode is their evaluation code, and the same four labels index the Hesse striations and an $A_2^4 \subset E_8$. On the committed executable E_8 Chevalley basis, a Hesse tick line induces the contact grading $1 + 56 + 134 + 56 + 1$; its 56 is exactly the Freudenthal triple system of the split Albert algebra whose norm is the repository’s signed 45-triad E_6 cubic, $Q = (\alpha\beta + T)^2 - 4\beta N(X) - 4\alpha N(Y) + 4T(X^\#, Y^\#)$, and the 57-dimensional Heisenberg extension carries the quasiconformal cone $N = Q/4 - \tau^2$ with the Günaydin–Koepsell–Nicolai inversion law. The Killing form obeys $K(e_\alpha, e_{-\alpha}) = -60(-1)^{\text{ht}\alpha}$; the three tick lines give commuting $E_{8(-24)}$ involutions with $E_8 = 80 + 56 + 56 + 56$, restricting to $E_{7(-25)}$, whose Hermitian Jordan triple yields a Euclidean Albert algebra with Peirce split $1 + 16 + 10$, a Lorentzian $(1,9)$ Peirce slice, $\text{Der} = F_{4(-52)}$, $\text{str}_0 = E_{6(-26)}$, and an $\mathfrak{so}(1,9)$ acting on a Majorana–Weyl 16. The repository’s matter-parity charge is $6L_c - 2$, and matter parity is the 2π rotation of $\text{Spin}(9)$. The committed split real form, by contrast, is the height twist $\exp(i\pi\rho^\vee)$ and gives a $(5,5)$ slice: signature is a real-form datum, invisible over $\mathbb{F}_3$. For the 215 heterotic $\mathbb{Z}_6$ Standard Models of the $W(3,3)$ twist we recover and freeze the exact spectra and prove, over the full rational $B - L$ freedom and over all 1024 $\mathbb{Z}_2$ characters odd on quarks and leptons, that no FI-cancelling D -flat singlet vacuum preserves matter parity: the never-even singlets have $B - L = \pm 1$ and every D -flat vacuum condenses one.

The forty points are forty views of one thing. Looked at through the right lens, the same finite object is a quantum measurement geometry, a finite space-time, a clock, a lattice, an exceptional Lie algebra, and a Lorentzian light cone — and each identification in this paper is an exact computation, not an analogy.

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How to read this paper

This paper is written for two readers at once. One has never studied algebra beyond school and wants to understand what has actually been found. The other wants every statement to be precise enough to check. We have tried not to cheat either of them. The mathematics is stated in full; next to it, in gold-edged boxes, the same idea is explained in ordinary words, and in teal-edged boxes we say what the idea means physically. Rose-edged boxes mark the edge of what is known.

The five badges

Every claim of substance carries one of five labels. They are the most important typographical device in the paper, because the whole value of a speculative subject lies in being able to tell the proved from the hoped-for.

THEOREM	A mathematical theorem: proved by an argument in the text, or a standard result with a reference.
COMPUTATION	An exact finite computation, carried out by a committed script whose output certificate is named. “Exact” means integer or rational arithmetic, or an exhaustive enumeration — never a numerical approximation, unless we say so.
BRIDGE	A bridge: a mathematically true statement <i>plus</i> an interpretation of it (for example, calling a mathematical light cone a light cone of time). The mathematics is certain; the interpretation is a proposal.
OPEN	An open question, stated as sharply as we can.
REFUTED	Something that was once believed, written down, and then shown false. We keep these, with the reason.

A map of the journey

The paper follows one chain of exact identifications. Figure 1 is the whole argument in one picture; each box is a section.

What you will need

Nothing beyond patience. We introduce every structure before we use it, and the glossary in Appendix B collects the vocabulary. Readers who want only the story can read the plain-words boxes and the key-idea panels and skip the displayed mathematics; nothing essential will be lost. Readers who want only the mathematics can do the reverse.

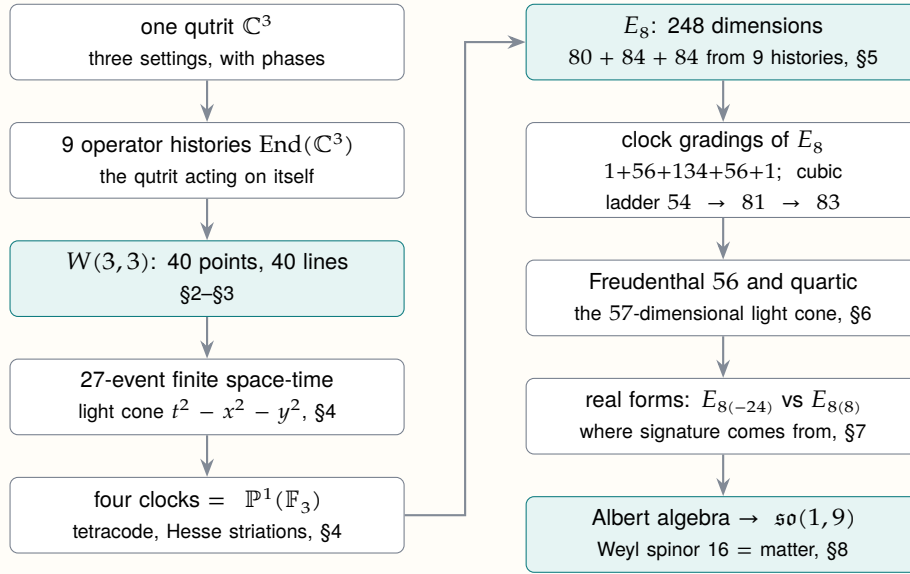


Figure 1. The chain of exact identifications followed in this paper. Shaded boxes are the three places where the story changes scale: from a qubit-sized object to a finite geometry, from the geometry to E_8 , and from E_8 to a Lorentzian algebra.

Where the material comes from

Everything here is drawn from the public repository `wilcompute/W33-Theory`, a record of several thousand numbered “passes” by two independent research tracks, and from its three long manuscripts — the $W(3,3)$ paper, the photonic Holonet paper and the Holonet machine blueprint — some of whose pages are adapted here. Passes 10949 and 10950, which supply Sections 6–8, were carried out for this synthesis. A reader who wants to check a statement can find its script and certificate in the ledger of Appendix A.

1 Prologue: a shape made of measurements

1.1 Why look at something finite?

Physics describes the world with continuous quantities: positions, times, field strengths, each a real number that can be divided forever. Yet the deepest facts we know about nature are *discrete*. An electron’s spin, measured along any axis, gives one of exactly two answers. Quarks come in exactly three colours. There are exactly three generations of matter. The symmetry groups that organise the forces — $SU(3)$, $SU(2)$, $U(1)$ — are continuous, but their representations, which decide what particles can exist, are labelled by whole numbers.

This programme starts from the discrete end. It asks: is there a single finite object, small enough to write down completely, whose structure already contains the patterns physics keeps finding? The candidate studied here is a configuration of forty points called $W(3,3)$. It is not new — geometers have known it for a century — but it has an unusual habit: every time it is examined through a different lens, it turns out to be a well-known structure from a different part of mathematics or physics. This paper is an account of those lenses, and of exactly how far each one reaches.

In plain words

A useful picture: $W(3,3)$ is like a crystal. A crystal is finite in its unit cell, but the unit cell determines the whole crystal's behaviour — how it breaks, how light passes through it, what sounds it makes. The hope of this programme is that $W(3,3)$ is a unit cell for something larger. The honest answer, as you will see, is that it is a remarkable unit cell for the *symmetries* of physics, and a much weaker one for its *dynamics*.

1.2 Bits, trits and qutrits

Every computer you have used stores information as bits — switches that are either off or on. A *trit* is a three-way switch: 0, 1, or 2. On its own that is not interesting; two bits can imitate a trit. A *qutrit* is different. It is a physical system with three distinguishable states, obeying quantum mechanics, which means it can be in a blend of all three settings at once, with relative phases. In symbols, a qutrit state is a unit vector

$$|\psi\rangle = a_0|0\rangle + a_1|1\rangle + a_2|2\rangle \in \mathbb{C}^3,$$

and the phases of the complex numbers a_j matter.

Two operations generate everything one can do to the “shape” of a qutrit. The *shift* X moves each setting to the next, and the *clock* Z stamps each setting with a phase:

$$X|j\rangle = |j+1 \bmod 3\rangle, \quad Z|j\rangle = \omega^j|j\rangle, \quad \omega = e^{2\pi i/3}.$$

They do not commute: $ZX = \omega XZ$. That single twist — a phase of one third of a turn picked up when the order of two operations is swapped — is the seed of everything in this paper.

What it means physically

X and Z are the finite versions of the two most important operations in quantum mechanics: translation in position and translation in momentum. The relation $ZX = \omega XZ$ is the three-state version of Heisenberg's $[x, p] = i\hbar$. Physicists call the group generated by X, Z and the phases the *Heisenberg–Weyl group*; its elements are the qutrit *Pauli operators*. They label the possible measurements one can make on a qutrit that are “as simple as possible” — the ones every other measurement is built from.

1.3 Arithmetic with three numbers

Because $X^3 = Z^3 = 1$, exponents only matter modulo 3. We therefore work in the *field with three elements*, $\mathbb{F}_3 = \{0, 1, 2\}$, where addition and multiplication are done modulo 3 — clock arithmetic on a three-hour clock. It is a genuine number system: every nonzero element has an inverse ($1 \cdot 1 = 1$, $2 \cdot 2 = 4 = 1$). In particular $2 = -1$ and $\frac{1}{2} = 2$, facts we will use repeatedly.

A qutrit Pauli operator is named by two numbers, $X^a Z^b$ with $(a, b) \in \mathbb{F}_3^2$. Two qutrits need four: $(a_1, b_1, a_2, b_2) \in \mathbb{F}_3^4$, a space with $3^4 = 81$ points.

1.4 The forty in one breath

Here is the whole construction, which Sections 2–3 unpack slowly. Label the 81 two-qutrit Pauli operators by vectors $x \in \mathbb{F}_3^4$. Remove the identity, leaving 80. An operator and its inverse describe the same measurement (one is the other run backwards), and x and $-x = 2x$ label that pair, so the distinct measurements number

$$\frac{3^4 - 1}{3 - 1} = \frac{80}{2} = 40.$$

Two of these forty measurements can be performed together exactly when their operators commute. Recording which pairs commute gives the forty-point geometry $W(3,3)$.

The forty points of $W(3,3)$ are the forty independent directions along which a pair of qutrits — or, as Section 3 shows, one qutrit acting on itself — can be measured. The lines of the geometry are the complete sets of measurements that can be made at the same time.

1.5 What this paper claims, and what it does not

A “theory of everything” should, at a minimum, reproduce the particles and forces of the Standard Model, the geometry of space-time and gravity, and the numbers — masses, couplings, mixing angles — that we measure. It is important to say at the start that nothing in this paper achieves that. What it does achieve is a tight web of *exact* connections between a single finite object and the structures that any such theory seems to need:

- the qutrit Heisenberg group and its Clifford symmetries (Sections 2–3);
- a finite space-time with a light cone, finite clocks, and a rigorous distinction between reversible time and an arrow of time (Section 4);
- the exceptional Lie algebra E_8 and its subalgebras E_7 and E_6 (Section 5);
- the Freudenthal triple system behind four-dimensional black-hole charges, and a 57-dimensional light cone on which E_8 acts by conformal-like maps (Section 6);
- the octonionic Albert algebra, the Lorentz algebra $\mathfrak{so}(1,9)$ and its chiral spinor, which carries exactly the quantum numbers of one generation of matter (Sections 7–8).

Section 9 is devoted to the physics that has been attempted on top of this — string compactifications that realise the Standard Model on the same structure — including the negative results, and Section 10 to a quantum computer architecture read off the same geometry. The Epilogue lists what remains open.

Where this stops

The finite geometry supplies *symmetry and structure*. It does not, by itself, supply a Hamiltonian, an energy scale, a value of Newton’s constant, or a reason for any particular mass. Where this paper uses words like “time”, “light cone” or “matter”, it is naming a mathematical structure that behaves like the physical one; whether nature uses it is a separate, empirical question.

2 The object: the symplectic quadrangle $W(3,3)$

2.1 Definition

Let $V = \mathbb{F}_3^4$ with coordinates $x = (x_1, x_2, x_3, x_4)$ and the *symplectic form*

$$\langle x, y \rangle = x_1y_3 + x_2y_4 - x_3y_1 - x_4y_2 \pmod{3}.$$

It is bilinear, alternating ($\langle x, x \rangle = 0$) and nondegenerate (only $x = 0$ pairs to zero with everything).

Definition 2.1. The *points* of $W(3,3)$ are the one-dimensional subspaces of V (40 of them). Two distinct points $\langle x \rangle, \langle y \rangle$ are *collinear* if $\langle x, y \rangle = 0$. The *lines* of $W(3,3)$ are the two-dimensional subspaces L on which the form vanishes identically (*totally isotropic* or *Lagrangian planes*); each contains $(9 - 1)/2 = 4$ points.

Proposition 2.1

THEOREM

$W(3,3)$ has 40 points and 40 lines. Every line has 4 points and every point lies on 4 lines. For a point p and a line L not containing it, exactly one point of L is collinear with p .

Proof. Points: $(3^4 - 1)/(3 - 1) = 40$. For a point $p = \langle x \rangle$, the set $x^\perp = \{y : \langle x, y \rangle = 0\}$ is a 3-dimensional subspace containing x ; the lines through p are the 2-dimensional subspaces of $x^\perp$ that contain x , and these correspond to the points of $x^\perp / \langle x \rangle \cong \mathbb{F}_3^2$, of which there are $(9 - 1)/2 = 4$. Double counting point–line incidences, $40 \cdot 4 = (\# \text{lines}) \cdot 4$, so there are 40 lines. Finally, if $p \notin L$ then $x^\perp \cap L$ is a subspace of L of dimension 1 (it cannot be all of L , since then $\langle x \rangle + L$ would be a totally isotropic 3-space, impossible in a nondegenerate 4-space), so exactly one point of L is collinear with p . $\square$

The last property is the defining axiom of a *generalized quadrangle*: there are no triangles made of three lines, but from any point you can always reach any line in one step, in exactly one way. With four points per line and four lines per point, $W(3,3)$ is the generalized quadrangle of order $(3,3)$, usually written $\text{GQ}(3,3)$ [31].

2.2 The collinearity graph and its spectrum

Join two points by an edge when they are collinear. Each point has $4 \times 3 = 12$ neighbours, so the graph has $40 \cdot 12/2 = 240$ edges. Two collinear points share exactly 2 further neighbours (the other two points of their line), and two non-collinear points share exactly 4 (the points of $x^\perp \cap y^\perp$, a nondegenerate plane). A graph in which these three numbers are constant is called *strongly regular*; this one is $\text{SRG}(40, 12, 2, 4)$ [5].

Proposition 2.2

THEOREM

The adjacency matrix A of $W(3,3)$ has eigenvalues 12 (once), 2 (24 times) and -4 (15 times).

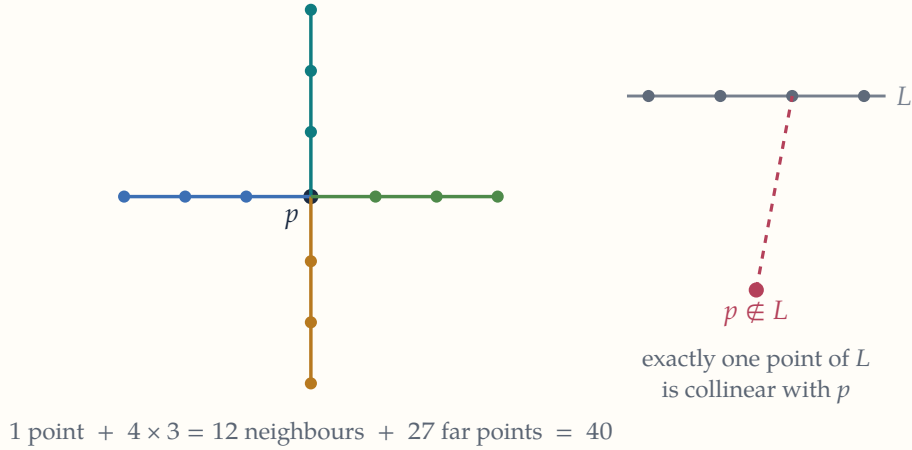


Figure 2. Left: the view of $W(3,3)$ from one point. Right: the quadrangle axiom.

Proof. For $\text{SRG}(v, k, \lambda, \mu)$ one has $A^2 = kI + \lambda A + \mu(J - I - A)$, where J is the all-ones matrix. On vectors orthogonal to the all-ones vector, J acts as zero, so every other eigenvalue θ satisfies $\theta^2 = (\lambda - \mu)\theta + (k - \mu)$, i.e. $\theta^2 + 2\theta - 8 = 0$, giving $\theta \in \{2, -4\}$. Multiplicities f, g satisfy $f + g = 39$ and $\text{tr } A = 12 + 2f - 4g = 0$, so $f = 24, g = 15$. $\square$

Since $|-4| < 2\sqrt{11} \approx 6.63$, the graph is *Ramanujan*: it is as good an expander as a 12-regular graph can be. This is why the machine of Section 10 can route between any two of its forty nodes in two hops, and why the graph's minimum bisection is large (100 of the 240 edges, a computed value).

In plain words

"Strongly regular" means that the geometry looks exactly the same from every point, and from every pair of points. There is no privileged measurement. The three eigenvalues are the geometry's "vibration frequencies"; that there are only three is the algebraic signature of this perfect evenness.

2.3 What the points and lines are physically

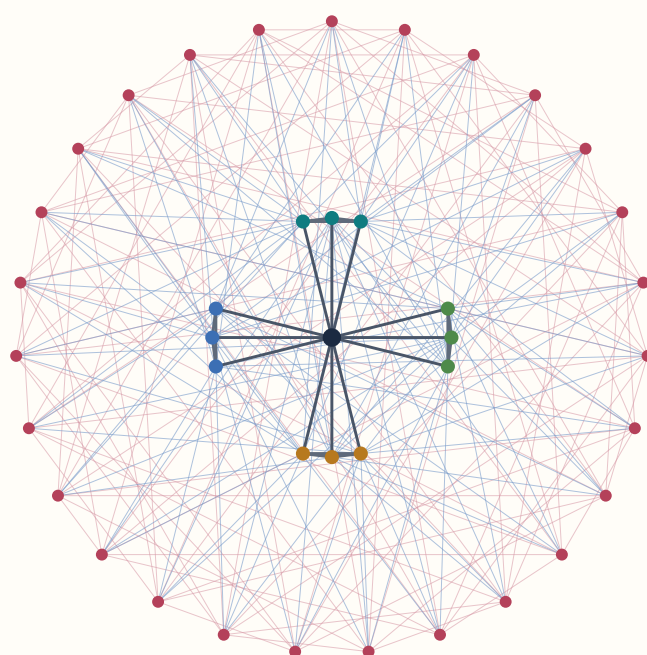
Let $W(x) = X^{x_1}Z^{x_3} \otimes X^{x_2}Z^{x_4}$ be the two-qutrit Pauli operator with label $x \in \mathbb{F}_3^4$. A direct calculation from $ZX = \omega XZ$ gives

$$W(x)W(y) = \omega^{\langle x, y \rangle} W(y)W(x). \quad (1)$$

So the symplectic form is not an arbitrary choice: it *is* the commutation phase of two-qutrit measurements.

What it means physically: contexts

- A **point** is a measurement direction: the cyclic group generated by one Pauli operator, which is the same whether it is run forwards or backwards.
- Two points are **collinear** when their operators commute and can be measured at the same time without disturbing each other.



$W(3,3)$: 40 points, all 240 edges

— p and its 4 lines (12 + 12 edges) — neighbour to far point (108) — far to far (108)

Figure 3. A portrait of $W(3,3)$, computed from the symplectic form and drawn from the point of view of one point p (centre). Its 12 neighbours lie on four lines (four colours, three points each); the 27 far points sit on the outer ring, ordered by which neighbour they see on each of the four lines — by the quadrangle axiom every far point sees exactly one. The edge counts $12 + 12 + 108 + 108 = 240$ are forced by $\text{SRG}(40, 12, 2, 4)$.

- A **line** is a maximal set of mutually compatible measurements: an abelian group of 9 Pauli operators whose common eigenvectors form a basis of the two-qutrit space $\mathbb{C}^9$. Physicists call this a *stabilizer group*, and each of its 9 joint eigenstates a *stabilizer state*. There are 40 lines and so $40 \times 9 = 360$ stabilizer states — the standard count for two qutrits.
- The **quadrangle axiom** says: given any measurement and any complete measurement context not containing it, exactly one member of the context is compatible with it.

The quantum operations that map Pauli operators to Pauli operators (up to phase) form the two-qutrit *Clifford group* [13]. Modulo the Paulis themselves and phases, it acts on the labels x by exactly the linear maps preserving $\langle, \rangle$: the symplectic group $\text{Sp}(4,3)$. So the symmetries of the geometry are, literally, the “cheap” quantum gates.

2.4 Two groups of order 51,840, which are not the same group

The symmetry groups that appear around $W(3,3)$ are easy to confuse, and confusing them has cost this programme real work (Section 5.4). Adapting the blueprint’s account:

$$\text{PSp}(4,3) \cong U_4(2) \text{ (order 25 920),} \quad \text{Sp}(4,3) \cong 2.U_4(2), \quad \text{PGSp}(4,3) \cong U_4(2):2 \cong W(E_6).$$

The last two both have order 51 840 but are *not isomorphic*: $\text{Sp}(4,3)$ has a centre of order two (the map $x \mapsto -x$), while $\text{PGSp}(4,3)$ — the full automorphism group of $W(3,3)$ — has trivial centre and is the Weyl group of the exceptional root system E_6 . **THEOREM**

In plain words

The two groups have the same size but different “shapes”. One of them contains a special element that commutes with everything — a global sign flip. Whether that sign is visible or invisible turns out to decide whether the structure can tell left from right. This is why the same forty points can *host* a handedness without being able to *choose* one (Section 9.5).

2.5 The dual picture and a firewall

Swap the roles of points and lines: the 40 lines of $W(3,3)$, with two lines “adjacent” when they meet, form the dual quadrangle, which is the parabolic quadric $Q(4,3)$. For odd q the quadrangle $W(3,q)$ is *not* isomorphic to its dual, although both have collinearity graph parameters $\text{SRG}(40, 12, 2, 4)$. This distinction matters several times below: the finite space-time of Section 4 lives on the *line* side, while the qutrit operators live on the *point* side. Repeated coincidences of counts between the two sides (for instance 27 vertices, 108 edges, 36 triangles) do not make them the same object — the repository calls this the *point/line firewall*, and it has caught several tempting mistakes. **COMPUTATION** (`w33_20260924_history_pointline_cospectral_firewall.py`)

2.6 Why three?

A sceptical reader should ask whether “three” was chosen because it works. The repository records five equations in which q appears as an unknown integer and each has exactly one positive solution, $q = 3$. Two can be checked by hand:

- The number of edges of the collinearity graph of $W(3, q)$ is $q(q+1)^2(q^2+1)/2$; the number of elements of $\mathbb{F}_{q^5}$ outside $\mathbb{F}_q$ is $q^5 - q$. Setting them equal gives $2(q-1) = q+1$, so $q = 3$: both equal 240.
- The number of points not collinear with a given point is q^3 ; it equals 27, the dimension of the fundamental representation of E_6 , only for $q = 3$.

Where this stops

These are coincidences of integers, not derivations. They do not prove that three is *necessary* for anything physical. They do show that the programme’s most unusual choice — three-level systems rather than two-level ones — is the one value at which several independent structures meet. **BRIDGE**

3 One qutrit, acting on itself

Section 2 built $W(3, 3)$ from *two* qutrits. This section shows that one is enough — provided we look not at the qutrit’s states but at the things that can be done to it. That shift of viewpoint is the precise content of the phrase the Holonet papers use, *self-entanglement*, and it is worth getting exactly right, because the phrase can also be read in ways that are not true.

3.1 Operators are states

An operator on one qutrit is a 3×3 matrix: nine complex numbers. So the space of operators $\text{End}(\mathbb{C}^3)$ is itself a nine-dimensional complex vector space, and with the Hilbert–Schmidt inner product $\langle A, B \rangle = \text{tr}(A^\dagger B)$ it is a Hilbert space. Writing $|A\rangle\rangle$ for a matrix regarded as a vector,

$$\text{End}(\mathbb{C}^3) \cong \mathbb{C}^3 \otimes \mathbb{C}^{3*} \cong \mathbb{C}^9, \quad |i\rangle\langle j| \longleftrightarrow |i\rangle \otimes |j\rangle.$$

Under this identification the identity matrix becomes the maximally entangled “Bell” state of two qutrits,

$$|\Omega\rangle = \frac{1}{\sqrt{3}}|I\rangle\rangle = \frac{1}{\sqrt{3}}(|00\rangle + |11\rangle + |22\rangle).$$

This is the Choi–Jamiołkowski correspondence, a standard tool of quantum information: every operation on a system is encoded by a state of two copies of it.

In plain words

Imagine describing a machine not by its current setting but by its complete instruction table: “if the input is j , produce i with amplitude A_{ij} ”. That table has nine entries. The nine entries behave exactly like the state of two three-way systems — an “input” and an “output”. Nothing mystical is happening; it is the same information written in a

different shape. The “two qutrits” of Section 2 can be read as the before-and-after of a single qutrit.

3.2 The two-sided action and its geometry

Let $P_u = X^{u_1} Z^{u_2}$ for $u \in \mathbb{F}_3^2$. A qutrit operator A can be multiplied by Pauli operators from the left and from the right:

$$\Phi_{u,v}(A) = P_u A P_v^\dagger, \quad (u, v) \in \mathbb{F}_3^2 \oplus \mathbb{F}_3^2.$$

Theorem 3.1

THEOREM

COMPUTATION

The 81 maps $\Phi_{u,v}$ form an orthogonal basis of $\text{End}(\text{End}(\mathbb{C}^3)) \cong M_9(\mathbb{C})$; the 80 non-identity maps span $\mathfrak{sl}_9$. They commute up to the phase

$$\Phi_{u,v} \Phi_{u',v'} = \omega^{[u,u'] - [v,v']} \Phi_{u',v'} \Phi_{u,v}, \quad [u, u'] = u_1 u'_2 - u_2 u'_1.$$

The form $[u, u'] - [v, v']$ is symplectic on $\mathbb{F}_3^4$, and the resulting commutation geometry on the 40 projective classes is $W(3, 3)$, with collinearity graph $\text{SRG}(40, 12, 2, 4)$ and spectrum $12^1 2^{24} (-4)^{15}$. `w33_20260924_temporal_su9_e8_compiler.py`

Proof of the phase. $\Phi_{u,v} \Phi_{u',v'}(A) = P_u P_{u'} A P_{v'}^\dagger P_v^\dagger$ and $\Phi_{u',v'} \Phi_{u,v}(A) = P_{u'} P_u A P_v^\dagger P_{v'}^\dagger$. Using $P_u P_{u'} = \omega^{[u,u']} P_{u'} P_u$ and taking adjoints of $P_v P_{v'} = \omega^{[v,v']} P_{v'} P_v$ gives the ratio $\omega^{[u,u'] - [v,v']}$. $\square$

The minus sign is the fingerprint of right multiplication: acting from the right reverses the order of products, so the right-hand copy of the qutrit carries the *opposite* algebra. The two “qutrits” are not two particles, and not even two subsystems in the usual sense: they are the algebra of left actions and its commutant, the algebra of right actions, on one qutrit’s operator space.

3.3 The Bell line is the set of physical channels

Among all $\Phi_{u,v}$, which ones are genuine quantum operations? A linear map on operators is physically realisable as a quantum channel only if it is *completely positive*, which for $\Phi_{u,v}$ means its Choi matrix $|P_u\rangle\rangle\langle\langle P_v|$ is positive semidefinite. A rank-one matrix $|a\rangle\langle b|$ is positive only when b is a positive multiple of a ; so up to phase, $\Phi_{u,v}$ is completely positive exactly when $u = v$. The diagonal $\Delta = \{(u, u)\}$ is totally isotropic ($[u, u'] - [u, u'] = 0$) and its four projective points form one line of $W(3, 3)$.

Proposition 3.2

THEOREM

COMPUTATION

The unitary channels $A \mapsto P_u A P_u^\dagger$ form a distinguished line of $W(3, 3)$, the *Bell line*; each fixes $|\Omega\rangle$. The left–right exchange $(u, v) \mapsto (v, u)$ reverses the symplectic form; its eight fixed rays split into two Lagrangian lines, and the $+$ line is exactly the unitary-

channel diagonal. `w33_20260924_liouville_swap_cp_outer_bridge.py`

What it means physically: a built-in time reversal

Exchanging left and right multiplication is the operator version of running a process backwards: $A \mapsto P_u A P_v^\dagger$ becomes $A \mapsto P_v A P_u^\dagger$. Because it reverses the sign of the commutation form, it is an *anti*-symmetry of $W(3,3)$ — the same kind of map as complex conjugation, which is how quantum mechanics implements time reversal. The repository shows that ordinary matrix transposition of the Pauli algebra induces exactly this class of outer symmetry, the one earlier work had identified abstractly (BT172). **COMPUTATION** (`w33_20260924_intrinsic_transpose_outer.py`, up to a labelling gauge.)

3.4 Three different things called “self-entanglement”

The phrase can mean three distinct things, and the Holonet programme has, at times, slid between them. It is worth separating them once and for all.

Notion	What is entangled	Status here
Intraparticle	Two independent degrees of freedom of one photon (for example its arrival-time bin and its frequency bin).	Standard and experimentally demonstrated.
Operator (Choi)	The input and output legs of one operation, as in §3.1.	The exact algebraic home of $W(3,3)$.
Temporal process	Quantum correlations between interventions at different times, described by a multi-time <i>process tensor</i> .	Needs its own witness (§3.5).

A correction the papers needed

An earlier draft of the photonic paper described three optical time bins as containing “a past qutrit and a future qutrit”. Three orthogonal bins span only $\mathbb{C}^3$, whereas two qutrits need $\mathbb{C}^9$; the two-bin subsets it named overlap and cannot be independent tensor factors. The architecture now uses two genuinely independent qutrit degrees of freedom of one photon — time bin $\otimes$ frequency bin — and reads “past” and “future” as the input and output legs of a process, not as simultaneously existing copies. **REFUTED** (the three-bin reading) **THEOREM** (the corrected encoding).

3.5 Memory across a causal break

What does it take for a system to carry quantum information from one time to a later one? The repository’s cleanest test is a three-step process with a qutrit memory M : swap the system S into M ; discard S and replace it with anything at all (a *causal break*); swap back. If the original state reappears, the process had memory. But a classical memory that merely

records “0, 1 or 2” would pass that test too. The decisive witness entangles S with a reference R first.

Theorem 3.3

COMPUTATION

With S initially maximally entangled with a reference qutrit R :

- a coherent memory returns the Bell state exactly (negativity 1, Bell fidelity 1);
- a classical label memory returns $\frac{1}{3} \sum_j |jj\rangle\langle jj|$ (negativity 0, Bell fidelity $\frac{1}{3}$), although it passes every basis-state retrieval test;
- a depolarizing memory $\rho \mapsto (1-p)\rho + pI/3$ returns an isotropic state with Bell fidelity $1 - \frac{8p}{9}$ and negativity $\max(0, \frac{3-4p}{3})$; quantum memory survives exactly while $p < \frac{3}{4}$.

w33_20260924_temporal_quantum_memory_witness.json

What it means physically

The threshold $p = \frac{3}{4}$ is the standard point at which a qutrit depolarizing channel becomes entanglement-breaking: beyond it, the link between the earlier and the later system can still carry classical information but no longer quantum information. Retrieval after a break proves that the past is *remembered*; preserved entanglement with a reference proves that it is remembered *quantumly*. Neither involves any influence travelling backwards in time.

3.6 A state picks a clock

A faithful qutrit state $\rho = \text{diag}(p_0, p_1, p_2)$ defines a natural flow on operators, the *modular flow* of Tomita–Takesaki theory, with generator $K = -\log \rho \otimes 1 + 1 \otimes \log \rho$ acting on $\text{End}(\mathbb{C}^3)$. The matrix units are its eigenvectors:

$$K E_{ij} = \log \frac{p_j}{p_i} E_{ij}.$$

The three diagonal units have frequency zero; the six off-diagonal units E_{ij} ($i \neq j$) — the six directed transitions between the three levels — come in three opposite pairs. Their pattern of differences is exactly the root system A_2 . **COMPUTATION** (w33_20260924_modular_a2_event_clock.py)

In plain words

A perfectly balanced qutrit ($p_0 = p_1 = p_2$) has no preferred direction: every transition has zero frequency, and nothing distinguishes “forward” from “backward”. Any imbalance gives each of the six transitions a frequency, and pairs each with its reverse at the opposite frequency. This is the finite form of an old idea of Connes and Rovelli [6]: that a state of a system can itself define a flow of time. It orients change; it does not by itself make change irreversible. That distinction — between an orientation of time and an arrow of time — returns in Section 4.

4 Time: a finite light cone and four clocks

Section 3 ended with a qutrit whose operators behave like the before-and-after of a process. This section asks what *time* looks like inside $W(3,3)$. The answer comes in three layers, and keeping them apart is the main lesson of the section:

1. a finite **present**: 27 “events” carrying an exact Minkowski-type form $t^2 - x^2 - y^2$, with a light cone and four null directions;
2. an unbounded **history**: the paths through those events, which never close up and which remember a winding number that the present forgets;
3. an **arrow**: the direction of time in the thermodynamic sense, which appears only when information is thrown away.

Everything in layers 1 and 2 is reversible. Layer 3 is where irreversibility enters, and we shall see exactly what it costs.

4.1 Choosing a “now”

Fix one line B of $W(3,3)$ — the Bell line of Section 3, the unitary channels. The other 39 lines fall into two kinds: 12 lines meet B (three through each of its four points), and 27 lines miss it entirely. Every line that misses B can be written uniquely as

$$L_S = \{ \langle Sy, y \rangle : y \in \mathbb{F}_3^2 \}, \quad S = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad a, b, c \in \mathbb{F}_3,$$

for a *symmetric* 2×2 matrix S . There are $3^3 = 27$ such matrices. So the lines of $W(3,3)$ decompose as

$$40 = \underbrace{1}_B + \underbrace{12}_{\text{meet } B} + \underbrace{27}_{\text{histories } S}.$$

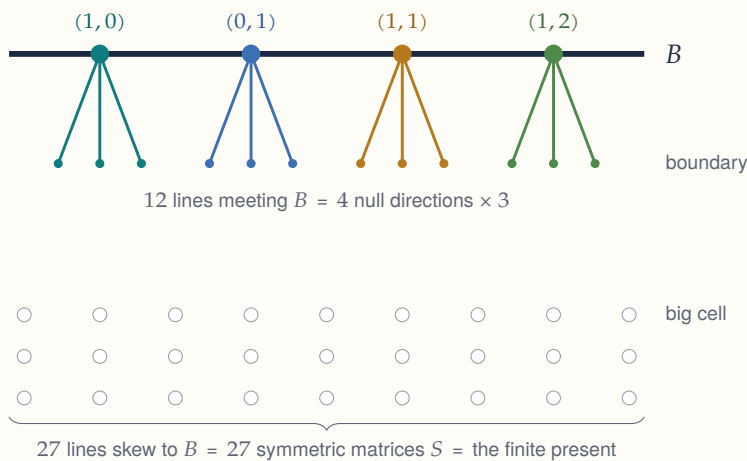


Figure 4. The forty lines of $W(3,3)$ seen from the Bell line B . The four points of B are labelled by the four projective directions of $\mathbb{F}_3^2$; they will turn out to be the four null directions of a finite light cone. The $1 + 12 = 13$ lines touching B form the “boundary”; the 27 others are the events of the finite present.

The 13 boundary lines are not an arbitrary leftover. In Plücker coordinates the 40 lines of $W(3,3)$ lie on the quadric $x_0x_4 - x_1x_3 + x_2^2 = 0$ in $\mathbb{P}^4(\mathbb{F}_3)$ — this is the dual quadrangle $Q(4,3)$ of Section 2 — and the 27 histories are its “big cell” $x_0 \neq 0$, reached by $S \mapsto [1 : a : b : c : ac - b^2]$. The 13 points with $x_0 = 0$ form the tangent cone of the quadric at B : B itself plus four three-point generators, one for each point of B . **COMPUTATION** (`w33_20260924_history_bigcell_q43_compactification.py`)

4.2 The finite light cone

Two histories L_S and L_T are either disjoint or they meet in a point. A short calculation shows they meet exactly when

$$\det(S - T) = 0.$$

Write $S = \begin{pmatrix} t+x & y \\ y & t-x \end{pmatrix}$ (possible because 2 is invertible mod 3). Then

$$\det S = t^2 - x^2 - y^2$$

— the Minkowski form of a space-time with one time and two space dimensions. Two events are *null-separated* (“light can travel between them”) exactly when their lines meet, i.e. when the two contexts share a measurement.

Theorem 4.1

THEOREM
COMPUTATION

The 27 histories form the finite Minkowski space $(\mathbb{F}_3^3, t^2 - x^2 - y^2)$. Relative to any event, the other 26 split as 8 null, 6 timelike ($\det = 1$) and 12 spacelike ($\det = -1$). The null directions through an event are the four rank-one matrices vv^T , $v \in \mathbb{P}^1(\mathbb{F}_3)$, and they correspond bijectively to the four points of B . `w33_20260924_history_bigcell_q43_compactification.py`

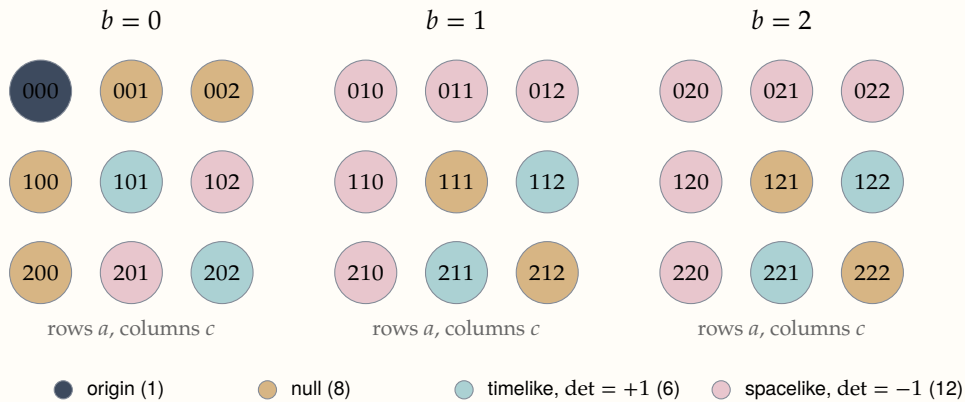


Figure 5. The 27 events $S = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, labelled abc and coloured by their separation from the origin 000 . The counts $1 + 8 + 6 + 12$ are those of a light cone over the field with three elements.

In plain words

In ordinary relativity, two events are “null-separated” if a flash of light could connect them. Here there is no light and no distance — only three possible values of each coordinate — yet the same algebraic form $t^2 - x^2 - y^2$ organises the 27 events, and

“null” has a sharp operational meaning: two complete measurement contexts share exactly one measurement. The *celestial circle* — the set of directions light can travel — is in ordinary $2 + 1$ dimensions a circle of directions; here it is the four points of the projective line over $\mathbb{F}_3$.

4.3 Four null pulses make one tick

The four null directions have integer representatives

$$D_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad D_\infty = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}, \quad D_+ = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \quad D_- = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix},$$

each positive semidefinite of rank one. Over $\mathbb{F}_3$ they satisfy a single relation, and it is worth staring at:

Proposition 4.2 (the common mode)

THEOREM

COMPUTATION

The counter map $\mathbb{F}_3^4 \rightarrow \text{Sym}_2(\mathbb{F}_3)$, $(n_0, n_\infty, n_+, n_-) \mapsto \sum n_i D_i$, is onto with kernel exactly $\langle (1, 1, 1, 1) \rangle$, so

$$\text{Sym}_2(\mathbb{F}_3) \cong \mathbb{F}_3^4 / \langle (1111) \rangle.$$

Over the integers, however,

$$D_0 + D_\infty + D_+ + D_- = 3I,$$

a strictly timelike matrix of trace 6. `w33_20260924_positive_null_counter_cover.py`

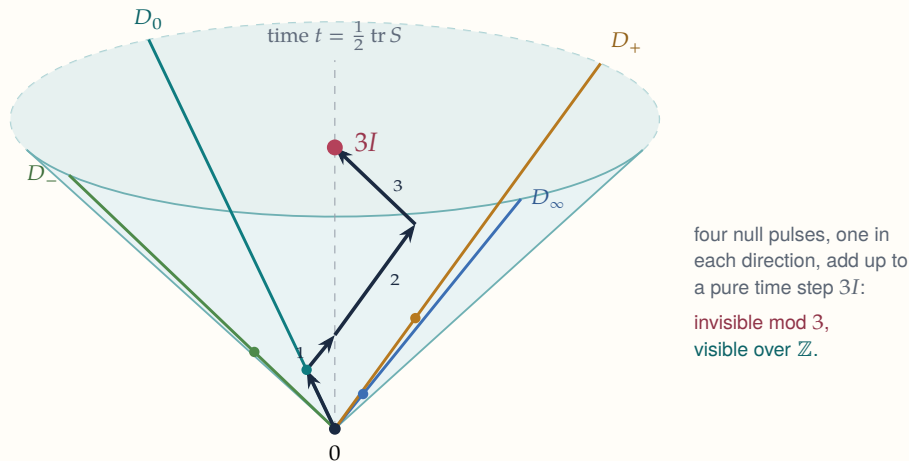


Figure 6. The real light cone of 2×2 symmetric matrices, drawn with time $t = \frac{1}{2} \text{tr } S$ upward. The four null rays $D_0, D_\infty, D_\pm$ are the lifts of the four points of the Bell line. The black zig-zag adds one pulse in each direction: it ends on the time axis at $3I$. Modulo three this is the zero matrix — the finite present cannot see it — but the integer lift records it as elapsed proper time.

What it means physically: two kinds of time

Proposition 4.2 separates two notions that ordinary physics merges. The *state* of the finite present is a counter of null pulses modulo their common mode: 27 values. But a history that fires one pulse in every direction returns to the same present while advancing by a pure time step upstairs. The finite chart measures *where* you are; the lifted history measures *how long* it took. This is the structural shape of the Page–Wootters idea that time is a relation between a clock and a system, and of the distinction, in relativity, between coordinate position and proper time along a world line. **BRIDGE**

4.4 Light-like steps: the null graph and its clock

Join two of the 27 events when they are null-separated. The resulting graph is 8-regular with 108 edges, and it can be diagonalised exactly.

Theorem 4.3 (spectral clock)

COMPUTATION

The null graph has adjacency spectrum $8^1 2^{12} (-1)^8 (-4)^6$ and Laplacian spectrum $0^1 6^{12} 9^8 12^6$. At $t^* = 2\pi/9$ the continuous-time walk $U = e^{-it^*L}$ satisfies $U^3 = I$, with eigenvalue multiplicities $1^9, \omega^{12}, (\omega^2)^6$. The nine fixed modes are the constant mode and the eight nonzero dual-null vectors of $\mathbb{F}_3^3$. `w33_20260924_null_history_spectral_clock.py`

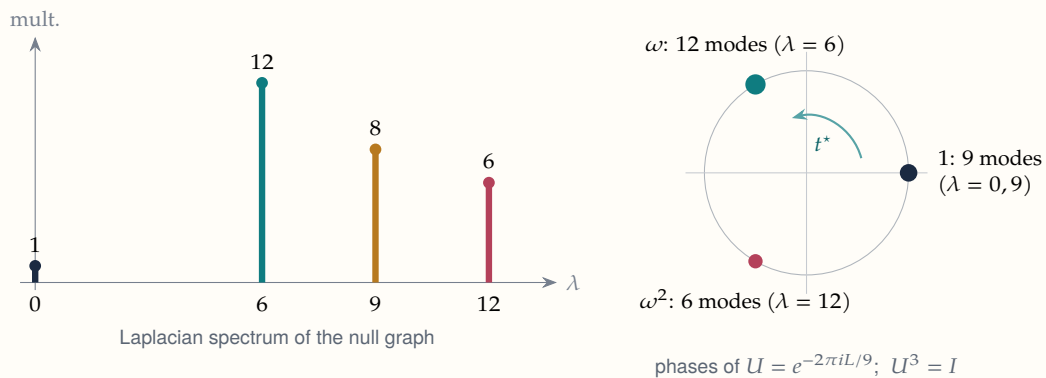


Figure 7. Left: the four Laplacian frequencies of the null graph. Right: running the graph as a quantum walk for time $2\pi/9$ sends every mode to a cube root of unity — the 27-event present carries an exact qutrit-valued clock.

A tempting identification that fails

The nine fixed modes invite the reading “9 = the nine operator histories of one qutrit” from Section 3. It is false. The symmetry group acting on the fixed nine is faithful of order 648, whereas the projective qutrit Clifford group has order 216, and the natural Bell-line S_4 splits the eight null modes $4 + 4$ while $SL(2, 3)$ is transitive on the eight nonidentity qutrit Pauli labels. The count matches; the module does not. **REFUTED** (as an identification)

4.5 History is a path, and paths do not close

The null graph is connected with 27 vertices and 108 edges, so its cycle space has rank $108 - 27 + 1 = 82$. Its fundamental group is the free group F_{82} and its universal cover is the infinite 8-regular tree.

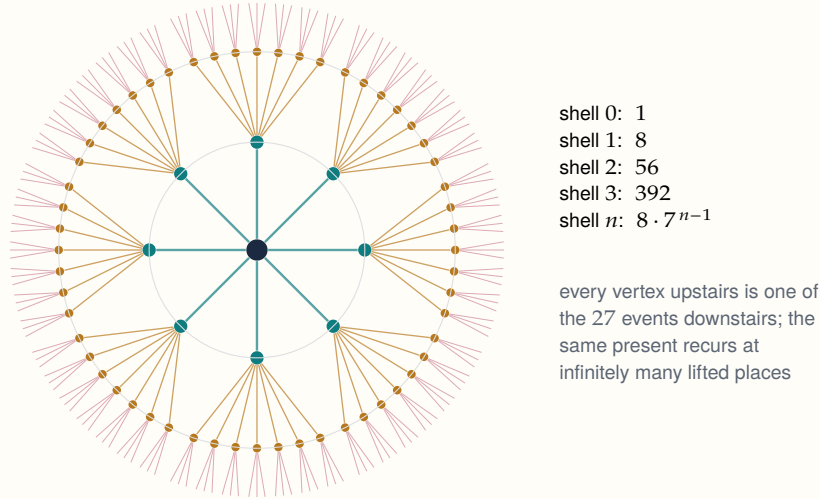


Figure 8. The universal cover of the null graph (only three grandchildren of each vertex are drawn in the outer shell). The present is finite; the history is an unbounded tree.

Theorem 4.4 (orientation and homology)

COMPUTATION

- (i) The 108 null edges are partitioned into 36 edge-disjoint triangles, one for each point of $W(3,3)$ off the Bell line.
- (ii) A signed sum of the 36 triangle boundaries is a primitive cycle through all 108 edges with coefficients ± 1 . The index-2 subgroup of the Bell stabiliser (order 648) fixes it, and the other coset negates it: the cycle space splits objectwise as $82 = 1_{\text{orientation}} + 81$.
- (iii) Filling the 36 triangles leaves first Betti number 46, and the reduced 81-dimensional module splits as $35 + 46$. The 46 injects, by an explicit chain map, into $H_1(W(3,3))$, which is itself 81-dimensional and irreducible — the Steinberg module of $\text{PSp}(4,3)$.

w33_20260924_history_invariant_cycle_orientation.py, w33_20260924_history_to_global_h1_intertwiner.py

So there is exactly one global orientation of the history graph, and the symmetry group of the “now” either preserves it or reverses it. This is a *direction* for time, but not yet an *arrow*: reversing it is a symmetry of the geometry.

Temporal observable	Lives on	Reversible?
position in the present	one of 27 events	yes
step count	length of a path in the null graph	yes
reduced word	element of the free group F_{82}	yes
winding	homology class in $\mathbb{Z}^{82}$; the common mode of Prop. 4.2	yes
orientation	the sign of the unique invariant cycle	yes (up to a symmetry)
coherent phase	the Weil phase μ_{12} below	yes (complex conjugation)
records	classical outcomes kept by an observer	no : discarding them is the arrow

4.6 A coherent twist: the twelfth roots of unity

For the qutrit Fourier transform F and the chirp $G = \text{diag}(1, \omega, \omega)$ one has $(FG)^3 = iF^2$. The metaplectic (Weil) representation that implements the symplectic symmetries on quantum states therefore carries phases in $\langle i, \omega \rangle = \mu_{12}$. Across the 27 quadratic forms of the present, the Weil phases occur with census $1^{13}, i^4, (-1)^6, (-i)^4$. Complex conjugation — the quantum-mechanical time reversal — sends $i \mapsto -i$ and $\omega \mapsto \omega^{-1}$. This is a reversible, coherent orientation mechanism, distinct from entropy production.

COMPUTATION

(w33_20260924_temporal_maslov_mul2.py)

4.7 Four clocks: the Hesse striations and the tetracode

The four points of the Bell line — the four null directions — reappear in a second, seemingly unrelated place. Identify the nine operator histories $|i\rangle\langle j|$ with the nine points of the affine plane $AG(2, 3) = \mathbb{F}_3^2$. Its 12 lines fall into four *parallel classes* of three lines each, one class for each direction $v \in \mathbb{P}^1(\mathbb{F}_3)$. Each class is a way of reading the nine cells as “three ticks of a clock”; we call the four classes the four *Hesse clocks*.

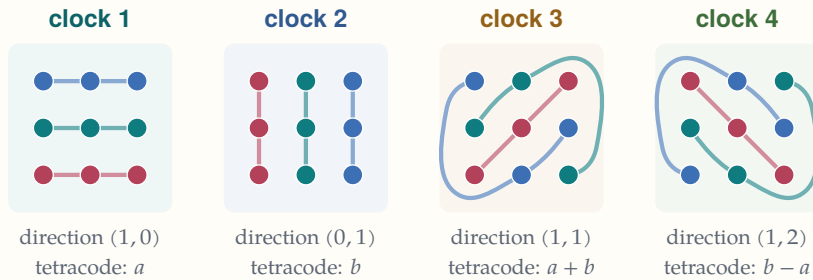


Figure 9. The four Hesse clocks. Each panel shows the nine operator histories of one qutrit as the affine plane $\mathbb{F}_3^2$, split into three parallel lines (the three “ticks”). Lines of the two diagonal classes wrap around, as on a torus. The four panels are labelled by the four points of $\mathbb{P}^1(\mathbb{F}_3)$ — the same four labels as the null directions of Figure 4 — and by the coordinate of the tetracode they read.

Theorem 4.5 (one S_4 for three structures)**COMPUTATION**

The four projective null directions of the present, the four Hesse clocks, and (Section 5) four orthogonal A_2 root systems inside E_8 are all indexed by $\mathbb{P}^1(\mathbb{F}_3) = \{(1,0), (0,1), (1,1), (1,2)\}$. For every one of the 24 elements of $PGL(2,3) \cong S_4$, the induced permutations of the three four-element sets are identical. The larger groups differ (the Bell stabiliser acts with kernel 27, $AGL(2,3)$ with kernel 18); the common object is the quotient S_4 and its action. `w33_20260924_null_hesse_4a2_s4_intertwiner.py`

The four clocks also carry a code. Evaluate the linear “clock reading” $\ell_{(a,b)}(x,y) = ax + by$ at oriented representatives $(1,0), (0,1), (1,1), (2,1)$ of the four directions. The result is

$$(a, b, a + b, b - a) \in \mathbb{F}_3^4,$$

and the nine such words are *exactly* the repository’s frozen *tetracode*, the $[4,2,3]_3$ code at the heart of the ternary Golay code and of the E_8 lattice’s tetracode construction — no coordinate permutation needed. **COMPUTATION** (`w33_pass10946_clock_code_cone_objectwise.py`)

(a,b)	$(1,0)$	$(0,1)$	$(1,1)$	$(2,1)$	
$(0,0)$	0	0	0	0	
$(1,0)$	1	0	1	2	
$(2,0)$	2	0	2	1	
$(0,1)$	0	1	1	1	
$(1,1)$	1	1	2	0	
$(2,1)$	2	1	0	2	
$(0,2)$	0	2	2	2	
$(1,2)$	1	2	0	1	
$(2,2)$	2	2	1	0	

nine clock readings $\ell_{(a,b)}$ evaluated
 at the four oriented null directions:
 every nonzero word vanishes on exactly one
 direction (one clock is “stopped”),
 minimum distance 3: the tetracode $[4,2,3]_3$.

Figure 10. The tetracode as a table of clock readings (colour: 1 teal, 2 rose).

What it means physically: a frame and an orientation

A projective direction picks a clock *frame*: which three-way split of the nine histories counts as the ticks. Choosing a representative of the direction — $(1,2)$ or its negative $(2,1)$ — picks an *orientation* of that clock. The tetracode’s peculiar sign $b-a$ (rather than $a-b$) in its last coordinate is exactly this orientation choice. Each nonzero codeword stops exactly one of the four clocks; 4 stopped clocks $\times$ 2 orientations $\times$ 3 origins give 24 complete clock calibrations.

The other research track found a further layer (Pass 10948, which we cite rather than re-derive). Over $\mathbb{F}_3$, $x^4 + 1 = (x^2 + x - 1)(x^2 - x - 1)$, and after flipping the orientation of one clock the tetracode becomes the ideal $(x^2 \pm x - 1)$ in $\mathbb{F}_3[x]/(x^4 + 1)$: the cyclic shift of clock readings with a sign, $N(c_0, c_1, c_2, c_3) = (-c_3, c_0, c_1, c_2)$, preserves it and satisfies $N^4 = -I$, $N^8 = I$. The two orientations give two transverse self-dual tetracodes forming a hyperbolic pair; the negacyclic presentation of the tetracode was used in a cyclotomic E_8 construction by Watanabe *et al.* [34]. **COMPUTATION** (`w33_pass10948_clock_tetracode_negacyclic_hyperbolic_bridge.py`) A clock whose fourth tick is a sign flip and whose eighth tick is

the identity behaves exactly like a spinor under rotation, and this is literal: the tick lifts to an element of order eight in $GL(2, 3)$ whose fourth power is the central 2π spin sign, and in Section 8 the same sign is matter parity (Theorem 8.3).

4.8 Clocks as relations: the Page–Wootters weld

Label a clock qutrit’s phase by $c \in \mathbb{F}_3$ and a system’s phase by $p \in \mathbb{F}_3$. The four directions of $\mathbb{F}_3^2$ are $c, p, c + p$ and $c - p$. For an order-three evolution U of the system, the two stationary “history states”

$$(X_{\text{clock}} \otimes U)|\Psi_+\rangle = |\Psi_+\rangle, \quad (X_{\text{clock}} \otimes U^{-1})|\Psi_-\rangle = |\Psi_-\rangle$$

are supported, in the clock’s phase basis, exactly on $c + p = 0$ and $c - p = 0$: the two correlated directions. The two remaining directions ($c = 0, p = 0$) are the uncorrelated axes. On the committed E_8 basis, pairs of axis-only backgrounds generate a subalgebra of dimension 24, whereas any graph-correlated background generates all 248 dimensions. **COMPUTATION**

(w33_20260924_page_wootters_diagonal_weld.py)

In plain words

In 1983 Don Page and William Wootters proposed [30] that the universe as a whole might be static, with time emerging as a correlation between a “clock” part and the rest. The finite version above is exact: a state that is globally unchanging, yet in which the clock reading and the system’s phase are locked together. Only the two *correlated* clock directions produce such states — and only they, in the E_8 computation, generate the whole algebra.

4.9 A second light cone: Hermitian $3 + 1$, and a weld that fails

Replace real symmetric matrices by Hermitian ones over $\mathbb{F}_9 = \mathbb{F}_3(u)$, $u^2 = -1$: $X = \begin{pmatrix} a & z \\ \bar{z} & c \end{pmatrix}$ with $z = x + uy$. Then $\det X = ac - x^2 - y^2$ is a form in four variables — a $3 + 1$ -dimensional finite Minkowski space.

Theorem 4.6 (the Hermitian present is a spread)

COMPUTATION

The 81 Hermitian events split $21 + 30 + 30$ by determinant, and the 40 projective directions split $10 + 15 + 15$. An explicit map sends the Hermitian quadric to the elliptic hyperplane section of $Q(4, 3)$, and under the line dictionary the ten null directions become ten pairwise disjoint lines of $W(3, 3)$ covering all 40 points — a *spread*. The 36 elliptic sections are exactly the 36 spreads of $W(3, 3)$. w33_20260924_hermitian_3p1_spread_bridge.py

The complex-structure weld does not exist

One might hope to weld the Hermitian light cone to the symplectic geometry through a single complex structure J ($J^2 = -I$) with $\Omega(x, y) = B(Jx, y)$. Exhausting all 729 candidates finds none. The light cone and the qutrit geometry meet through the point/-line duality of Section 2, not through one compatible J . **REFUTED** (for the direct weld)

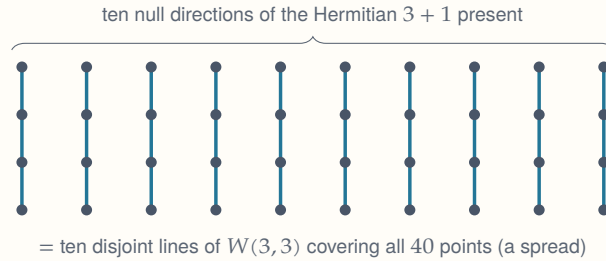


Figure 11. The Hermitian 3 + 1 light cone is a partition of the forty points into ten complete measurement contexts — a complete “schedule” for measuring every two-qutrit direction exactly once.

4.10 The arrow: forgetting two trits

Every structure so far is reversible. Where does irreversibility come from? There is a sharp answer.

Theorem 4.7 (the price of an arrow)

THEOREM

For every qutrit state ρ ,

$$\frac{1}{9} \sum_{u \in \mathbb{F}_3^2} P_u \rho P_u^\dagger = \text{tr}(\rho) \frac{I}{3}.$$

Each map $\rho \mapsto P_u \rho P_u^\dagger$ is unitary and reversible; their average, obtained by forgetting which u occurred, is the completely depolarising channel, which erases every state to the same fixed point. The forgotten record is two trits, so by Landauer’s principle [24] making a qutrit clock irreversible in this way dissipates at least $2 k_B T \ln 3$ per step.

Proof. The left side, as a function of ρ , commutes with conjugation by every P_v (which merely permutes the u ’s up to phases that cancel). Its image Σ therefore commutes with all nine Paulis, which span $\text{End}(\mathbb{C}^3)$, so Σ is a scalar; taking traces gives the scalar $\text{tr}(\rho)/3$. $\square$

What it means physically

The finite present has a light cone; the history has an orientation; the Weil phase has a handedness. None of these is an arrow of time, because each can be reversed by a symmetry. The arrow appears exactly when a record is discarded. For a qutrit clock the minimal record is the two-trit label of the Pauli step — the photonic papers’ “two trits per cycle” (`bt823_the_closure.py`; see also `w33_20260924_adqc_record_arrow.py`) — and discarding it turns a reversible tick into the maximally irreversible channel. This is the finite form of the modern view that the thermodynamic arrow is information loss, and it is *consistent* with the Connes–Rovelli idea of Section 3: a state can orient time, but only forgetting makes it flow one way. **BRIDGE**

Three layers of time, exactly separated: a finite present with a light cone (27 events, $t^2 - x^2 - y^2$); an unbounded, oriented history (a free group, an 8-regular tree, a winding the present cannot see); and an arrow that costs exactly two trits of forgotten record per tick.

Where this stops

None of this derives a continuum space-time, a speed of light, a metric that curves, or gravity. The light cone here is an algebraic form over a three-element field; “null” means “shares a measurement”. Whether the present, history and arrow of the finite model are the finite shadows of physical time is a bridge hypothesis, and the dynamics that would make it testable — a Hamiltonian, an energy scale — is not supplied by the geometry. [OPEN](#)

5 E_8 from nine histories

E_8 is the largest of the five exceptional simple Lie algebras: 248 dimensions, 240 roots, a symmetry so rigid that it appears, often unexpectedly, in string theory, in the densest packing of spheres in eight dimensions, and in the physics of certain magnetic materials. This section shows that one qutrit’s nine operator histories build E_8 in a way in which every piece has a meaning in the finite geometry — and it records carefully one famous coincidence that turned out to be a trap.

In plain words

A Lie algebra is the “infinitesimal” version of a continuous symmetry: the set of all directions in which you can nudge a system while keeping its rules intact, together with the rule (the *bracket*) for what happens when two nudges are combined in opposite orders. Its *roots* are the special directions that the symmetry’s own “clock” (its maximal commuting part) rotates at definite rates. E_8 has 240 of them.

5.1 Eighty plus eighty-four plus eighty-four

Recall from Section 3 that the 80 non-identity two-sided Weyl maps on one qutrit span $\mathfrak{sl}_9$, the Lie algebra of traceless 9×9 matrices acting on the nine operator histories $|i\rangle\langle j|$. Now add *packets*: unordered triples of histories. There are $\binom{9}{3} = 84$ of them, forming the representation $\Lambda^3 \mathbb{C}^9$; their duals form $\Lambda^6 \mathbb{C}^9$, another 84. A classical construction [1, 2] makes

$$E_8 = \mathfrak{sl}_9 \oplus \Lambda^3 \mathbb{C}^9 \oplus \Lambda^6 \mathbb{C}^9, \quad 248 = 80 + 84 + 84,$$

into a Lie algebra. On the nine history cells the brackets are literal packet operations: E_{ij} replaces cell j by cell i inside a packet; two disjoint triples bracket to the complementary six-set; a triple and a six-set differing by one cell contract to an E_{ij} ; opposite packets return a Cartan direction. [COMPUTATION](#) (`w33_20260924_temporal_su9_e8_compiler.py`, `w33_20260924_temporal_a8_e8_process_bracket.py`)

5.2 Hesse lines are copies of A_2

Place the nine histories on the affine plane $\mathbb{F}_3^2$ as in Figure 9. A triple S of cells labels the root

$$w_S = \sum_{i \in S} e_i - \frac{1}{3} \sum_{i=1}^9 e_i, \quad w_S \cdot w_T = |S \cap T| - 1.$$

(Indeed $|w_S|^2 = 3 - 2 + 1 = 2$.) The 84 triples fall into two orbits of the affine group $AGL(2, 3)$ (order 432): the 12 *lines* of the plane and the 72 non-collinear *triangles*.

Theorem 5.1 (four clocks, four A_2 's)

THEOREM

COMPUTATION

The three lines of one parallel class give three roots with pairwise products -1 and sum 0; with their negatives they form a root system A_2 . Lines of different classes meet in exactly one cell, so their roots are orthogonal. The 12 Hesse lines therefore span a reflection-closed subsystem $4A_2 \subset E_8$, one A_2 per clock, and the root shell refines as

$$240 = \underbrace{72}_{A_8} + \underbrace{24}_{\text{Hesse } 4A_2} + \underbrace{144}_{\text{triangles}}.$$

w33_20260924_temporal_hesse_4a2_a8_bridge.py

Proof. Disjoint triples have $|S \cap T| = 0$, hence product -1 ; the three lines of a class partition the nine cells, so $\sum w_S = \sum_i e_i - 3 \cdot \frac{1}{3} \sum_i e_i = 0$. Two non-parallel lines of an affine plane meet in one point, giving product $1 - 1 = 0$. $\square$

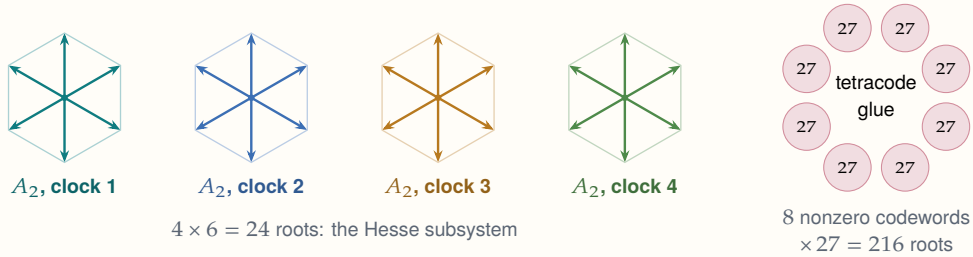


Figure 12. E_8 assembled from four clocks. Each Hesse clock contributes one A_2 root system (left). The remaining 216 roots come in eight sectors of 27, one for each nonzero word of the tetracode (right): $240 = 4 \cdot 6 + 8 \cdot 27$.

5.3 The glue is the tetracode, and the tetracode is the clock

The lattice E_8 contains the sublattice A_2^4 with quotient $\mathbb{F}_3^2$, and the well-known construction of E_8 from the tetracode [7] reads it as

$$E_8 = \bigcup_{c \in \mathcal{T}} (c + A_2^4), \quad 240 = 4 \cdot 6 + 8 \cdot 27,$$

where $\mathcal{T} \subset \mathbb{F}_3^4$ is the tetracode: each of the eight nonzero codewords has weight three, and contributes $3^3 = 27$ minimal vectors. Combined with Section 4, the four glue coordinates are the four clocks and the eight nonzero sectors are the eight oriented, nonzero linear clock readings. **THEOREM** (classical) **COMPUTATION** (the clock reading, w33_pass10946_clock_code_cone_objectwise.py)

5.4 A trap: $240 = 240$

The collinearity graph of $W(3, 3)$ has 240 edges, and E_8 has 240 roots. For years the repository looked for a natural bijection between them. The question has a precise and negative answer.

Theorem 5.2 (no equivariant bijection)

COMPUTATION

The two groups of order 51 840 of Section 2 each do only half the job:

- $\text{PGSp}(4, 3) \cong W(E_6)$ acts faithfully and transitively on the 240 edges of $W(3, 3)$, but *intransitively* on the 240 roots of E_8 ;
- $\text{Sp}(4, 3)$ acts faithfully and transitively on the 240 roots, but its centre acts trivially on the edges, so its edge action is not faithful.

Hence no bijection between edges and roots is equivariant for a common group. `w33_pass1020_e8_transitive_51840.g`

What *does* exist is better. The Weyl group of E_8 contains a regular element w of order three. By Springer's theory [32] its centraliser is the complex reflection group $G_{32} \cong \mathbb{Z}_3 \times \text{Sp}(4, 3)$ of order 155 520 — the symmetry group of the *Witting polytope*, whose 240 vertices in $\mathbb{C}^4$ are the E_8 roots.

Theorem 5.3 (the 6 : 1 fibration)

COMPUTATION

The 240 roots of E_8 fall into 40 complex lines $\{\pm v, \pm \omega v, \pm \omega^2 v\}$, each a copy of the six Eisenstein units. $\text{Sp}(4, 3)$ permutes the 40 lines with subdegrees 1, 12, 27: the quotient is the collinearity graph of $W(3, 3)$ — on its *points*, not its lines. The fibre group is $\langle c^5 \rangle \cong \mathbb{Z}_6$ for a Coxeter element c of $W(E_8)$. `w33_pass1021_e8_fibration_over_forty.g`

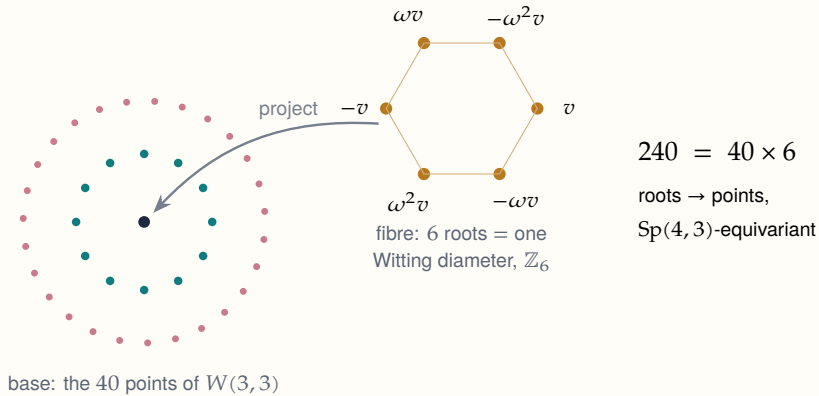


Figure 13. The correct relation between E_8 and $W(3, 3)$: not edges to roots, but a six-to-one map from roots to points, whose fibres are the Eisenstein unit hexagons.

What it means physically: a photon's four-dimensional state space

The 40 Witting diameters are 40 rays in $\mathbb{C}^4$ — the state space of a photon's path and polarisation, or of two qubits. Two rays are orthogonal exactly when the corresponding points of $W(3, 3)$ are collinear. So the same geometry is drawn twice: by *states* of a four-level system (the Witting rays) and by *operators* on two qutrits (the Pauli classes). The

photonic Holonet papers call this operator–operand duality. **COMPUTATION** (bt817_self_entangled_photon_atlas.py, bt821_operator_operand_duality.py)

5.5 $E_6 \times A_2$, the twenty-seven, and a cubic

Under its subalgebra $E_6 \oplus A_2$,

$$E_8 = (78, 1) \oplus (1, 8) \oplus (27, 3) \oplus (\overline{27}, \overline{3}),$$

which is a $\mathbb{Z}_3$ -grading $248 = 86 + 81 + 81$. The 27 of E_6 carries a unique invariant cubic form N ; in the repository’s committed basis it has exactly 45 signed terms (*triads*), each coordinate in 5 triads and each pair in at most one — precisely the incidence of the 27 lines and 45 tritangent planes of a smooth cubic surface, whose symmetry group is $W(E_6)$, the automorphism group of $W(3, 3)$ [12]. **THEOREM** (classical) **COMPUTATION** (canonical_su3_gauge_and_cubic.json)

What it means physically

In grand unified theories built on E_6 , the 27 holds one generation of matter: under $SO(10)$ it splits $27 = 16 + 10 + 1$ — sixteen fermions (including a right-handed neutrino), a ten of Higgs-type fields, and a singlet. The cubic N is then the unique renormalisable Yukawa coupling 27^3 . Section 8 shows this same split arising as scalar, vector and spinor of a ten-dimensional Lorentz group.

5.6 Clock gradings of E_8

A clock gives more than a $\mathbb{Z}_3$ -grading: once an origin and an orientation are chosen it gives an *integer* grading. On the committed Chevalley basis:

Theorem 5.4 (Hesse clocks grade E_8)

COMPUTATION

- (i) A single Hesse line (a “tick”) induces the *contact grading*

$$E_8 = \mathfrak{g}_{-2} \oplus \mathfrak{g}_{-1} \oplus \mathfrak{g}_0 \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2, \quad 1 + 56 + 134 + 56 + 1,$$

with $\mathfrak{g}_0 = E_7 \oplus \mathbb{C}$.

- (ii) An oriented striation (three parallel ticks) induces the *|3|-grading*

$$2 + 27 + 54 + 82 + 54 + 27 + 2,$$

with Levi $\mathfrak{g}_0 = E_6 \oplus A_1 \oplus \mathbb{C}$, $\mathfrak{g}_{\pm 1} = 27 \otimes 2$, $\mathfrak{g}_{\pm 2} = 27$, $\mathfrak{g}_{\pm 3} = 2$.

- (iii) The repository’s frozen $\mathbb{Z}_3$ -grading $86 + 81 + 81$ is exactly this integer grading reduced modulo three; all 8 347 nonzero committed structure constants obey the integer degree law.

w33_20260925_hesse_clock_e8_gradings.py, w33_20260925_e8_parabolic_cubic_clock_lift.py

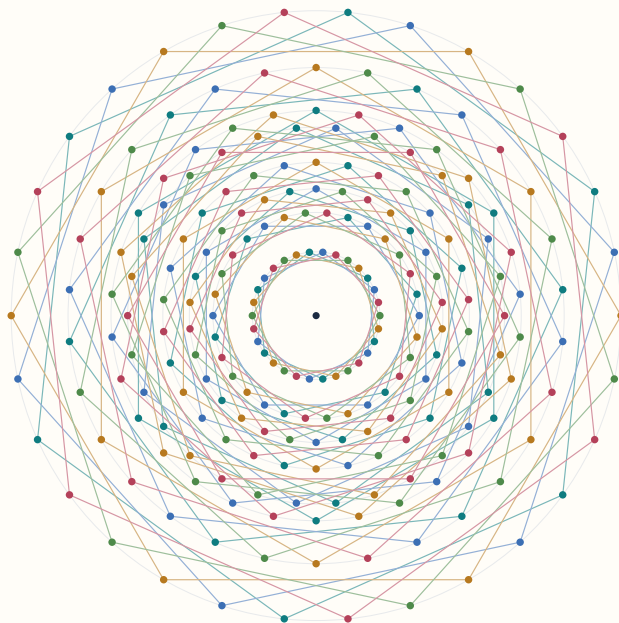


Figure 14. The 240 roots of E_8 projected to the Coxeter plane of a Coxeter element c (order 30), which acts on the plane as a rotation by 12° . The roots lie on eight rings of thirty. The orbits of c^5 — rotation by 60° — are regular hexagons, five per ring: these 40 hexagons are the 40 fibres of Theorem 5.3, one for each point of $W(3,3)$. Each hexagon is $\{\pm v, \pm \omega v, \pm \omega^2 v\}$, since $c^{15} = -1$ and c^{10} has order three. All coordinates computed (`make_figures.py`).

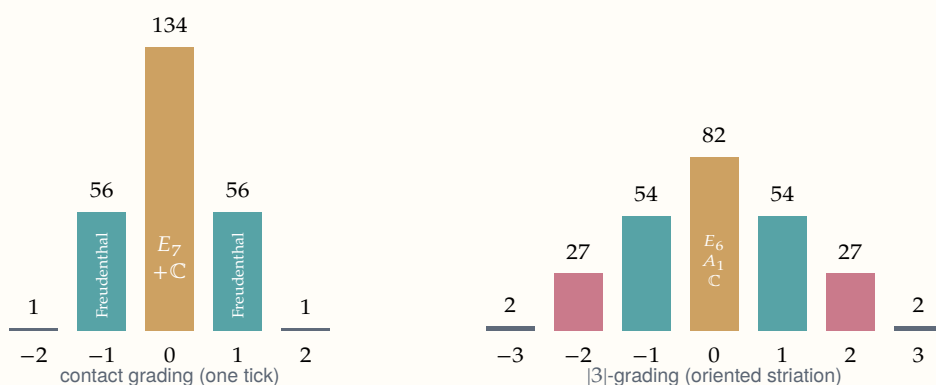


Figure 15. Two ways a clock slices E_8 . Heights are dimensions. On the left, the Freudenthal “56” of Section 6 appears at degree ± 1 ; on the right, the 27 of E_6 appears doubled at degree ± 1 and single at degree ± 2 .

This parabolic spectrum was found independently by Kraft, Regeta and Zimmermann in a study of small varieties with group actions [21]. What is repository-specific is the identification with the Hesse clocks and the committed structure constants.

5.7 The cubic ladder: three steps to the top

In the $|3|$ -grading, positive degrees form a nilpotent algebra $\mathfrak{g}_+ = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \mathfrak{g}_3$ of dimension $54 + 27 + 2 = 83$. It is generated by $\mathfrak{g}_1$, and the way it fills up is governed by the E_6 cubic.

Theorem 5.5 (the bracket ladder)

COMPUTATION

On the committed Chevalley bracket:

- (i) $[\mathfrak{g}_1, \mathfrak{g}_1] = \mathfrak{g}_2$: 270 nonzero pairs hit all 27 grade-two roots, each ten times;
- (ii) $[\mathfrak{g}_1, \mathfrak{g}_2] = \mathfrak{g}_3$: 54 nonzero pairs hit both top roots, each 27 times;
- (iii) the nonzero nested triple brackets exhibit exactly the 45 triads of the cubic, each 12 times, and for every signed triad d_{ijk} , $[[e_{(i,0)}, e_{(j,1)}], e_{(k,a)}] = -d_{ijk} T_a$ for $a \in \{0, 1\}$.

So the positive part grows $54 \rightarrow 81 \rightarrow 83$, and the committed cubic is exactly the rule by which three first-order moves reach the two-dimensional top. `w33_20260925_cubic_clock_bracket_ladder.py`

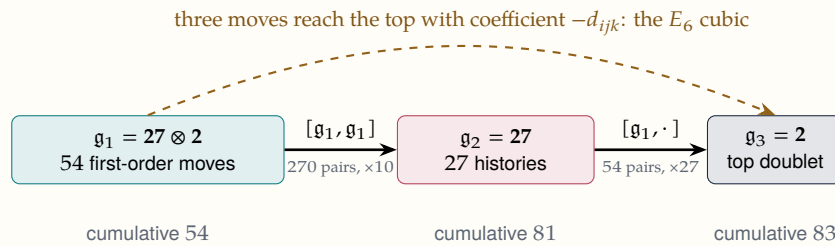


Figure 16. The cubic ladder. First-order moves combine in pairs to histories, and histories combine with one more move to reach the two-dimensional top. The only way three moves reach the top is through a triad of the cubic.

What it means physically: cubic time

Read degree as “order in time”. A first-order move is a state update; two moves compose into a second-order *history*; and three reach the top doublet, whose coefficient is the cubic volume N spanned by the three moves. In this reading the E_6 cubic — in GUT language the Yukawa coupling — is the rule by which three independent updates produce a net displacement of the clock. The algebra is exact; the reading is a bridge, and it lacks what would make it physics: a Hamiltonian, a state and an energy scale. **BRIDGE**

One qutrit's nine histories build E_8 as $80 + 84 + 84$. The four Hesse clocks are four orthogonal A_2 's whose glue is the tetracode; a clock tick grades E_8 as $1 + 56 + 134 + 56 + 1$, an oriented clock as $2 + 27 + 54 + 82 + 54 + 27 + 2$; and E_8 's roots fibre six-to-one over the forty points.

6 The Freudenthal 56 and a 57-dimensional light cone

The contact grading of Theorem 5.4 has a 56-dimensional layer $\mathfrak{g}_{-1}$. In the 1950s Hans Freudenthal showed [11] that the 56-dimensional representation of E_7 carries a remarkable structure — a symplectic form and a quartic form — now called a *Freudenthal triple system*. It has since turned up in the physics of black holes in supergravity and in the entanglement of qubits. This section computes it directly from the Hesse clock, on the committed E_8 basis, and finds a light cone one dimension larger.

In plain words

Think of the 56 as the space of “charges” of a black hole in a theory with many electric and magnetic fields: 28 electric and 28 magnetic. The symplectic form pairs each electric charge with its magnetic partner. The quartic form is a single number built from all 56 charges; in string theory its square root gives the black hole’s entropy. What we show is that the forty-point geometry, through its clock, produces exactly this structure — including the right signs, with nothing adjusted by hand.

6.1 Two clocks in one table

Let θ be the highest root of E_8 and α_8 the simple root with $\theta = \omega_8$. The tick line’s contact grading is the α_8 -coefficient a_8 of a root; the striation’s |3|-grading is its α_7 -coefficient a_7 . Counting roots by both at once gives Figure 17.

$a_7 =$	-3	-2	-1	0	1	2	3	
$a_8 = -2$	1	.	.	.	.	.	.	
$a_8 = -1$	1	27	27	1	.	.	.	contact layer $\mathfrak{g}_{-1}$ $1 + 27 + 27 + 1 = 56$
$a_8 = 0$	.	.	27	72 (+8 Cartan)	27	.	.	clock layer $27 + 27 = 54$
$a_8 = 1$	.	.	.	1	27	27	1	overlap: one 27
$a_8 = 2$	.	.	.	.	.	.	1	

Figure 17. The 240 roots of E_8 counted by contact degree a_8 (rows) and clock degree a_7 (columns). The Freudenthal 56 (solid outline) and the clock’s first-order 54 (dashed outline) share exactly one 27.

Theorem 6.1 (the 56 does not extend the 54)

COMPUTATION

The contact layer $a_8 = -1$ sits at clock degrees 0, -1, -2, -3 with sizes 1, 27, 27, 1; the clock layer $a_7 = -1$ sits at contact degrees 0, -1 with 27 each. The two share exactly one 27. The second 27 of the Freudenthal system is the clock’s *second-order* layer, and its two scalar “poles” are $\alpha = e_{-\alpha_8}$ (clock degree 0) and $\beta = e_{-(\theta-\alpha_8)}$ (clock degree -3, a member of the clock’s top doublet). `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`

A count that misled

A proposal circulating in the programme read the 56 as “the clock’s 54 plus the 2 top directions”. The dimensions agree ($54+2 = 56$), but the objects do not: the 56 contains only half of the 54. **REFUTED** (as an identification)

6.2 The symplectic form and the Heisenberg algebra

Brackets of two elements of the 56 land in the one-dimensional $\mathfrak{g}_{-2} = \mathbb{C}e_{-\theta}$: $[x_i, x_j] = \Omega_{ij} e_{-\theta}$. On the root basis, Ω is a signed perfect matching: $\alpha \leftrightarrow \beta$, and $X_k \leftrightarrow Y_k$ where X_k and Y_k carry the *same* 27-label. Its determinant is ± 1 , so it is unimodular over $\mathbb{Z}$. Since $e_{-\theta}$ is central in $\mathfrak{g}_{-1} \oplus \mathfrak{g}_{-2}$, the two together form a 57-dimensional *Heisenberg algebra* — the same algebra that, in quantum mechanics, is generated by positions, momenta and $i\hbar$. **COMPUTATION**

6.3 The quartic, in Freudenthal’s normal form

Define $Q(x) = \frac{1}{6}$ coefficient of $e_{-\theta}$ in $\text{ad}_x^4 e_\theta$ for $x \in \mathfrak{g}_{-1}$.

Theorem 6.2 (the tick line is a Freudenthal triple system)**COMPUTATION**

Q is an integral polynomial with 1036 monomials (primitive coefficients $\{1 : 28, \pm 2 : 378, \pm 4 : 630\}$), exactly invariant under all 126 root generators of the Levi E_7 . Writing $x = (\alpha, X, Y, \beta)$ with X, Y in the two 27’s,

$$Q = (\alpha\beta + T(X, Y))^2 - 4\beta N(X) - 4\alpha N(Y) + 4T(X^\#, Y^\#)$$

where N is the repository’s committed 45-triad E_6 cubic, $X_k^\# = \partial N / \partial X_k$, and $T(X, Y) = \sum_k t_k X_k Y_k$ with $t_k = \Omega(X_k, Y_k) \Omega(\alpha, \beta)$. No sign has been adjusted by hand. Moreover $(X^\#)^\# = N(X) X$ and $X \cdot X^\# = 3N(X)$, so N is the norm of a cubic Jordan algebra (a split Albert algebra) over $\mathbb{Q}$ [22]. `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`

The theorem makes one structural point vivid: the cubic “history volume” $N(X)$ of Section 5 enters the quartic *only* multiplied by a clock pole β .

Black holes and qubits

In $\mathcal{N} = 8$ supergravity the 56 electric and magnetic charges of a black hole transform under $E_{7(7)}$, and the Bekenstein–Hawking entropy is $S = \pi \sqrt{|Q|}$ with Q Cartan’s quartic invariant [20]; in the “magic” $\mathcal{N} = 2$ supergravity built on the octonionic Albert algebra the same formula holds with $E_{7(-25)}$. Duff and Ferrara showed that the same quartic measures the tripartite entanglement of seven qubits [9], and Borsten, Duff and collaborators built a dictionary between black holes and qubit entanglement on it [4]. Theorem 6.2 places that quartic, with its signs, inside the forty-point geometry’s clock. The two real forms named here — $E_{7(7)}$ and $E_{7(-25)}$ — are exactly the two that appear, for two different reasons, in Section 7. **BRIDGE**

6.4 Jordan frames are the forty-five triads

A *frame* of a cubic Jordan algebra is a decomposition of the identity into three orthogonal idempotents. On the committed basis, in each of four 27-layers, the 45 triads of the cubic are exactly the 45 strongly orthogonal triples of roots: the coordinate frames. (This corrects an older reading of “27 minimal idempotents”: 27 counts the coordinate rank-one directions; the frames are the 45 triads.) For every coordinate idempotent, the Peirce decomposition is $27 = 1 + 16 + 10$. **COMPUTATION**

6.5 The 57-dimensional light cone

Now use the 57-dimensional Heisenberg algebra as a space of points $p = (X, \tau)$, with $X \in \mathfrak{g}_{-1}$ and τ the central coordinate, and move the top vector e_θ around: $v_p = \exp(\text{ad}(X + \tau e_{-\theta})) e_\theta$.

Theorem 6.3 (quasiconformal light cone)

COMPUTATION

On the committed basis, symbolically:

- (i) the $e_{-\theta}$ -component of v_p is $\mathcal{N}(X, \tau) = \frac{1}{4}Q(X) - \tau^2$;
- (ii) the points compose by the Heisenberg law $(X_1, \tau_1)(X_2, \tau_2) = (X_1 + X_2, \tau_1 + \tau_2 + \frac{1}{2}\Omega(X_1, X_2))$;
- (iii) the *separation* $D(p, p') = \mathcal{N}(X - X', \tau - \tau' - \frac{1}{2}\Omega(X', X))$ equals $B(v_p, v_{p'})/B(e_{-\theta}, e_\theta)$ for the Killing form B ;
- (iv) the Weyl reflection w exchanging e_θ and $-e_{-\theta}$ acts as an *inversion* ι , and

$$D(\iota p, \iota p') = \frac{D(p, p')}{\mathcal{N}(p) \mathcal{N}(p')}.$$

This is the Günaydin–Koepsell–Nicolai quasiconformal realisation of E_8 [14], now computed objectwise on the repository’s basis. `w33_pass10949_freudenthal_quasiconformal_clock_cone.py`

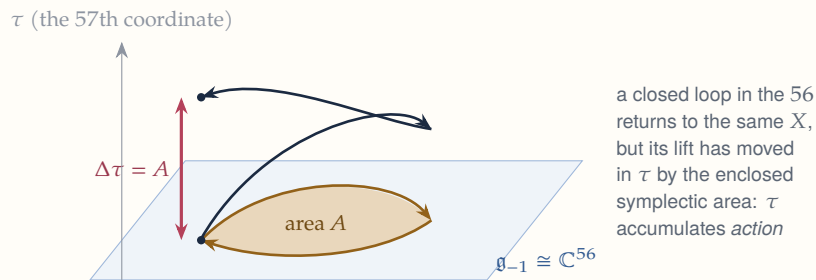


Figure 18. Why the 57th coordinate matters. In a Heisenberg group the central coordinate of a lifted path grows by the symplectic area its projection encloses. The quasiconformal separation D depends on $\tau - \tau'$ only through this corrected difference $\tau - \tau' - \frac{1}{2}\Omega(X', X)$.

What it means physically: a light cone, and its inversions

Ordinary Minkowski space has a *conformal* symmetry group, $SO(4, 2)$, larger than the Lorentz group; its most striking element is the inversion $x \mapsto x/x^2$, under which the interval between two events transforms as

$$(x' - y')^2 = \frac{(x - y)^2}{x^2 y^2}.$$

Theorem 6.3(iv) is exactly this law, with the Minkowski interval replaced by the quartic separation D on a 57-dimensional space. So E_8 acts on the forty-point clock's Heisenberg algebra as a group of “conformal” symmetries of a quartic light cone. The central coordinate τ plays the role of accumulated action: the same pattern met in Section 4, where the finite present forgot a common mode that the lifted history remembered. **BRIDGE** (the reading) **COMPUTATION** (the formulas)

A single tick of a Hesse clock turns E_8 into $1+56+134+56+1$. The 56 is Freudenthal's triple system, whose quartic — computed from the committed brackets with no sign adjusted — is the black-hole and seven-qubit invariant; and with one more coordinate it becomes a 57-dimensional light cone on which E_8 acts by conformal-type inversions.

Where this stops

The formulas are exact, but they are the formulas of the *complexified* (or split) structure. Whether the light cone is Lorentzian — one time direction, the rest space — depends on a choice of real form, which the finite field cannot see. That is the subject of Section 7. Nothing here supplies a black hole, a charge lattice of a physical theory, or dynamics.

OPEN

7 Real forms: where time's signature comes from

Every computation so far could have been done over the complex numbers — or over $\mathbb{F}_3$. But physics needs more: it needs to know which directions are *time-like* and which *space-like*. Mathematically, that is the choice of a *real form*.

In plain words

A complex Lie algebra is like a map drawn without saying which way is north. A real form chooses, for every direction, whether it behaves like a rotation (you come back to where you started — “compact”, like space) or like a boost (you never come back — “non-compact”, like the relation between observers moving at different speeds). The same complex E_8 has three real forms, and they differ precisely in how many directions are rotation-like.

7.1 Three real forms of E_8

A real form is fixed by a *Cartan involution*, which splits the algebra into a compact part $\mathfrak{k}$ (rotations) and a non-compact part $\mathfrak{p}$ (boosts). The difference $\dim \mathfrak{p} - \dim \mathfrak{k}$ is the *signature*.

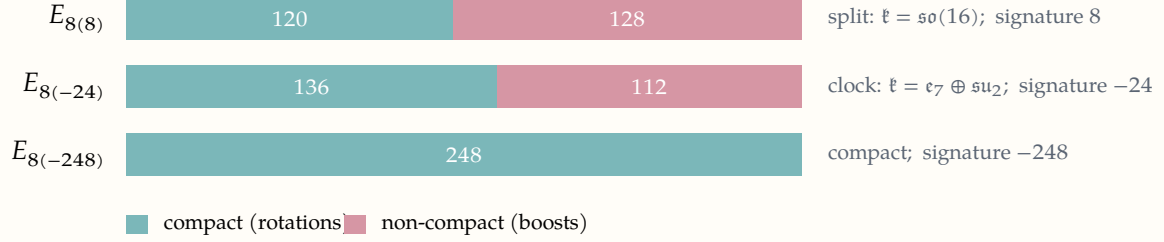


Figure 19. The three real forms of E_8 , drawn as the split $248 = \dim \mathfrak{k} + \dim \mathfrak{p}$. The committed “split” form of the repository is $E_{8(8)}$; the Hesse clocks select $E_{8(-24)}$.

7.2 A sign law, and the height twist

On the committed Chevalley basis the Killing form obeys an exact law,

$$K(e_\alpha, e_{-\alpha}) = -60 (-1)^{\text{ht } \alpha},$$

where $\text{ht } \alpha$ is the height of the root (the sum of its simple-root coefficients). From it one reads off the *compact* conjugation $\sigma_c(e_\alpha) = (-1)^{\text{ht } \alpha} \overline{e_{-\alpha}}$, verified to be an automorphism on all 30 628 basis pairs. The repository’s certified *split* involution Θ turns out to be σ_c composed with the *height parity* $(-1)^{\text{ht}} = \exp(i\pi\rho^\vee)$, where $\rho^\vee$ is the Weyl co-vector. **COMPUTATION** (w33_pass10949_freudenthal_quasiconformal_clock_cone.py; prior certificate w33_e8_split_real_form_involution.py)

7.3 The three tick lines are a Klein four-group

The three roots θ , $-(\theta - \alpha_8)$ and $-\alpha_8$ that define the tick line and its companions sum to zero, have pairwise products -1 , and are orthogonal to E_6 : they form the *external* A_2 of $E_6 \times A_2$. Each defines a parity: $(-1)^{a_8}$, $(-1)^{a_7}$, $(-1)^{a_7+a_8}$.

Theorem 7.1 (tick lines give $E_{8(-24)}$)

COMPUTATION

Composed with σ_c , each of the three tick-line parities defines the real form $E_{8(-24)}$. The three commute and form a Klein four-group whose simultaneous eigenspaces decompose

$$E_8 = \underbrace{80}_{\mathfrak{e}_6 \oplus \mathfrak{t}_2} + 56 + 56 + 56,$$

one Freudenthal 56 per tick line. w33_pass10949_freudenthal_quasiconformal_clock_cone.py

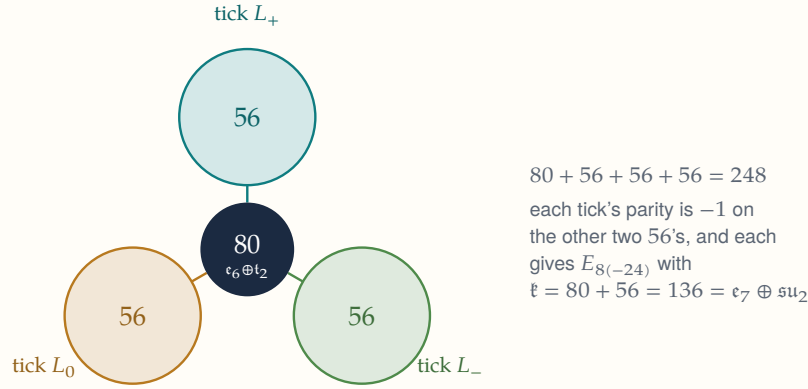


Figure 20. The three tick lines of a Hesse clock, and the decomposition of E_8 under the Klein four-group of their parities.

7.4 What each twist does to E_7 and E_6

twist of σ_c	E_8	contact Levi E_7	E_6
tick L_+ : $(-1)^{a_8}$	$E_{8(-24)}$	compact	compact
tick L_0 : $(-1)^{a_7}$	$E_{8(-24)}$	$E_{7(-25)}$	compact
tick L_- : $(-1)^{a_7+a_8}$	$E_{8(-24)}$	$E_{7(-25)}$	compact
height $(-1)^{\text{ht}}$ (committed split)	$E_{8(8)}$	$E_{7(7)}$	$E_{6(2)}$
height $\cdot L_+$ / height $\cdot L_0$	$E_{8(8)}$	$E_{7(7)} / E_{7(-5)}$	$E_{6(2)}$
height $\cdot L_-$	$E_{8(-24)}$	$E_{7(-5)}$	$E_{6(2)}$

Two supergravities in one table

$E_{7(7)}$ is the hidden symmetry (“U-duality”) of maximal $\mathcal{N} = 8$ supergravity in four dimensions, and $E_{8(8)}$ that of maximal supergravity in three. $E_{7(-25)}$ is the duality group of the “magic” $\mathcal{N} = 2$ supergravity built on the octonions. The committed split form of the repository lands on the first; the Hesse clock lands on the second. The black-hole quartic of Section 6 is the entropy formula in both.

More precisely, the clock’s chain $E_{6(-26)} \subset E_{7(-25)} \subset E_{8(-24)}$ (all three computed here, the first in Section 8) is exactly the chain of duality groups of the octonionic magic $\mathcal{N} = 2$ supergravity of Günaydin, Sierra and Townsend in five, four and three space-time dimensions [15]: $E_{6(-26)}$ acts on the 27 charges in five dimensions, $E_{7(-25)}$ on the 56 in four, and $E_{8(-24)}$ is the symmetry of the three-dimensional theory obtained by one more reduction. The split chain plays the same role for maximal supergravity.

BRIDGE

7.5 The Lorentzian slice lives only in the clock real form

The contact Levi $E_{7(-25)}$ has a Hermitian symmetric structure: on the 27-dimensional space $\mathfrak{p}^+ = \{a_8 = 0, a_7 = +1\}$ the triple product

$$\{x, y, z\} = -[[x, \sigma_c y], z]$$

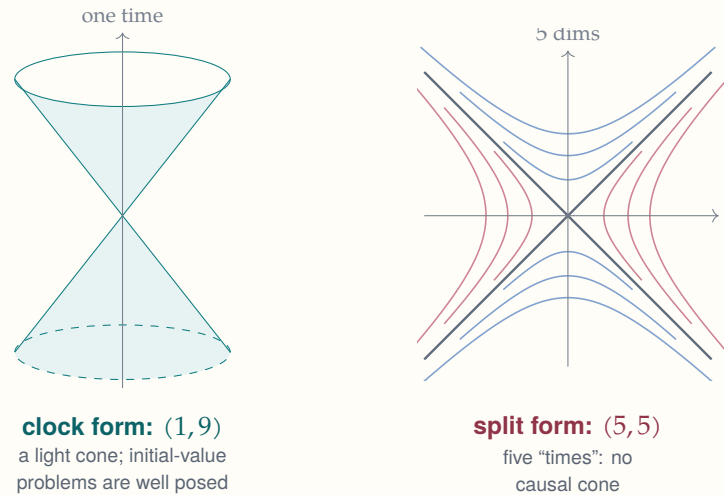
makes $\mathfrak{p}^+$ a Hermitian Jordan triple system. A *tripotent* is an element with $\{e, e, e\} = 2e$ (the normalisation used here).

Theorem 7.2 (signature is a real-form datum)

COMPUTATION

- (i) Under the clock conjugation all 27 root vectors of $\mathfrak{p}^+$ are tripotents; under the committed split conjugation only 12 of 27 are.
- (ii) For the tripotent e of a triad frame, $A = \{x : \{e, x, e\} = 2x\}$ with $x \circ y = \frac{1}{2}\{x, e, y\}$ is a 27-dimensional commutative Jordan algebra with *positive-definite* trace form, inertia $(27, 0, 0)$: a Euclidean Jordan algebra of rank three — the Albert algebra $H_3(\mathbb{O})$ [10].
- (iii) For a primitive idempotent of the frame, the Peirce spaces have dimensions $1 + 16 + 10$, and the rank-two determinant on the 10-dimensional Peirce-0 space has inertia $(1, 9)$ — Lorentzian.
- (iv) In the committed split form the same construction gives inertia $(5, 5)$. Over $\mathbb{F}_3$ the two quadrics are both of type $O^+(10, 3)$ and cannot be told apart.

w33_pass10949_freudenthal_quasiconformal_clock_cone.py



over $\mathbb{F}_3$: both are $O^+(10, 3)$ — indistinguishable

Figure 21. The same Peirce-0 quadric in two real forms of E_8 . Only the real form selected by the Hesse clock is Lorentzian.

What it means physically: one time, not five

Tegmark observed that only space-times with exactly one time direction support predictable physics: with several, the equations of motion are ultrahyperbolic and initial data do not determine the future [33]. Theorem 7.2 says the finite geometry does not decide between one time and five — over $\mathbb{F}_3$ they are the same — but that the real form picked out by its own clock is the one-time, Lorentzian choice. The committed split form, chosen earlier for computational convenience, is the other one.

BRIDGE

Signature is not visible over a finite field. It is a choice of real form, and the Hesse clock's tick lines make that choice: they select $E_{8(-24)} \supset E_{7(-25)}$, and with it a Euclidean Albert algebra whose Peirce slice is a $1 + 9$ light cone.

8 The Albert algebra: ten dimensions, a spinor, and matter parity

Section 7 produced, from the clock, a 27-dimensional Euclidean Jordan algebra. By the Jordan–von Neumann–Wigner classification there is exactly one such algebra that is “exceptional”: the *Albert algebra* $H_3(\mathbb{O})$ of 3×3 Hermitian matrices over the octonions. Since the 1930s it has been suspected of having something to do with quantum mechanics and particle physics; this section computes its symmetries on the committed basis and finds the Lorentz group of ten-dimensional space-time, a chiral spinor, and the repository’s own matter parity.

In plain words

The *octonions* are the largest of the four number systems in which you can divide: the real numbers, the complex numbers, the quaternions, and the octonions, of dimensions 1, 2, 4, 8. Octonions do not even obey $(ab)c = a(bc)$, yet 3×3 “Hermitian” matrices of them — real numbers on the diagonal, octonions above it, their conjugates below — form a consistent algebra of dimension $3 + 3 \times 8 = 27$. Its symmetries are the exceptional groups F_4 and E_6 [2]. Manogue, Dray and Wilson have recently described the Standard Model through octonionic E_8 structures of this kind [27].

8.1 Anatomy of a 3×3 octonionic matrix

Choose the idempotent $c = E_{11}$ (a 1 in the top-left corner). Every element of the algebra splits by how it behaves under multiplication by c — its *Peirce decomposition*, with eigenvalues $1, \frac{1}{2}, 0$:

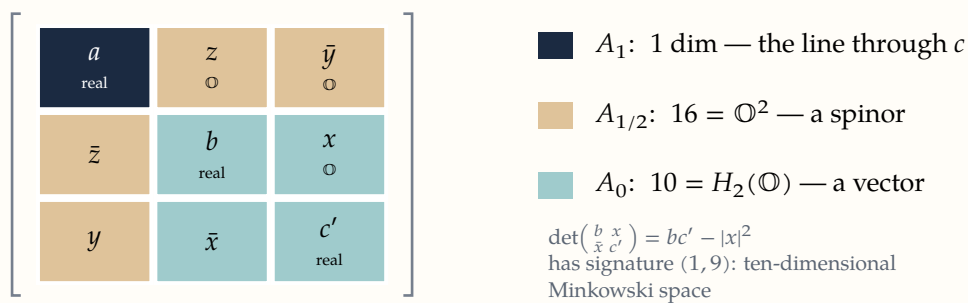


Figure 22. The Peirce decomposition $27 = 1 + 16 + 10$ of the Albert algebra with respect to a primitive idempotent. The lower-right 2×2 block is a copy of ten-dimensional Minkowski space; the two octonions in the first row and column form a sixteen-component spinor.

That the 2×2 Hermitian octonionic matrices form ten-dimensional Minkowski space, with the determinant as the Minkowski interval, is a classical observation (Sudbery; Kugo and

Townsend; Baez and Huerta), and it is the algebraic reason the superstring lives in ten dimensions [3, 23]. The new content is that the forty-point clock reaches it, on the committed basis.

8.2 The symmetries, computed

Theorem 8.1 (the clock Albert algebra)

COMPUTATION

For the 27-dimensional clock algebra A of Theorem 7.2:

- (i) the derivations $\text{Der}(A) = \text{span}[L_x, L_y]$ form a 52-dimensional Lie algebra with negative-definite trace form $(0, 52)$: the compact $F_{4(-52)}$;
- (ii) the reduced structure algebra $\text{str}_0(A) = \text{Der}(A) \oplus L(A_{\text{tr}=0})$ is 78-dimensional, obeys $[D, L_x] = L_{Dx}$ for all 52×27 pairs, and has inertia $(26, 52)$: $E_{6(-26)}$;
- (iii) the derivations fixing c form a compact 36-dimensional $\mathfrak{spin}(9)$; adding the nine “boosts” L_y ($y \in A_0$, orthogonal to $e - c$) gives a 45-dimensional algebra that preserves A_0 and its determinant of inertia $(1, 9)$, and acts faithfully: it is $\mathfrak{so}(1, 9)$;
- (iv) on $A_{1/2}$ the commutant of $\mathfrak{so}(1, 9)$ is one-dimensional: the 16 is absolutely irreducible — the Majorana–Weyl spinor of $\mathfrak{so}(1, 9)$.

w33_pass10950_clock_albert_lorentz_spinor.py

	ordinary space-time $H_2(\mathbb{C})$: 4 dims	superstring space-time $H_2(\mathbb{O})$: 10 dims	exceptional space-time $H_3(\mathbb{O})$: 27 dims
rotations (automorphisms)	$SO(3)$	$\text{Spin}(9)$ ✓	$F_{4(-52)}$ ✓
Lorentz (norm-preserving)	$SO(1, 3)$	$\text{Spin}(1, 9)$ ✓	$E_{6(-26)}$ ✓
conformal	$SO(2, 4)$	$SO(2, 10)$	$E_{7(-25)}$ ✓
quasiconformal	—	—	$E_{8(-24)}$ ✓

✓ computed in this paper on the committed E_8 basis, in the real form selected by the Hesse clock (Passes 10949–10950); unmarked cells are classical.

Figure 23. The ladder of space-times. A Jordan algebra of Hermitian matrices is a space-time; its determinant is the interval. For 2×2 complex matrices it is ordinary Minkowski space, for 2×2 octonionic matrices the ten-dimensional space of the superstring, and for 3×3 octonionic matrices the 27-dimensional “exceptional space-time” of Günaydin [16]. Each has rotations, a Lorentz group, a conformal group and (for the last) a quasiconformal group. The Hesse clock’s real form of E_8 produces the whole right-hand column, and the middle column inside it.

What it means physically: the clock picks the division algebra

The last row of Freudenthal’s “magic square” of Lie algebras can be built from the octonions in two ways: from the division algebra $\mathbb{O}$, giving the chain $F_{4(-52)} \subset E_{6(-26)} \subset E_{7(-25)} \subset E_{8(-24)}$ of automorphism, Lorentz, conformal and quasiconformal groups of $H_3(\mathbb{O})$, or from the split octonions, giving the split forms. The Hesse clock selects

the first: the *division-algebra* octonions, the ones with a positive norm, the ones that make $H_2(\mathbb{O})$ a genuine Minkowski space. The repository's earlier "split" choice lands on the other side. **COMPUTATION** (the four real forms) **BRIDGE** (the reading)

8.3 Down to four dimensions

Choose a further splitting of the 10 into 4 + 6: time $u = e - c$, one frame direction, and a complex line inside the octonions supplied by the complex structure of $\mathfrak{p}^+$ itself. The subalgebra of $\mathfrak{so}(1, 9)$ preserving the 4 and the 6 is 21-dimensional, $\mathfrak{so}(1, 3) \oplus \mathfrak{so}(6) \cong \mathfrak{sl}(2, \mathbb{C}) \oplus \mathfrak{su}(4)$, and its commutant on the 16 is two-dimensional and contains a complex structure. Hence

$$16 = (2, 4) \oplus \overline{(2, 4)}.$$

COMPUTATION (w33_pass10950_clock_albert_lorentz_spinor.py)

Two readings of the same branching

A ten-dimensional chiral spinor, seen by a four-dimensional observer, becomes four left-handed Weyl spinors transforming as the 4 of $SU(4)$. If $SU(4)$ is read as the rotation group of six hidden dimensions, this is the fermion content of maximally supersymmetric Yang–Mills theory. If it is read as Pati–Salam colour, with lepton number as the fourth colour, it is one generation of quarks and leptons. The branching is exact; which reading nature uses is not decided by it. **BRIDGE**

8.4 Matter parity is a rotation by 2π

The repository carries an independent certificate for a "matter charge" Q_ψ on the 27: it splits $27 = 16_1 + 10_{-2} + 1_4$, and on the 45 triads the charges fall into the pattern $40 \times (-2, 1, 1) + 5 \times (-2, -2, 4)$. Matter parity is $(-1)^{Q_\psi}$: odd on the 16, even on the 10 and 1. In the Minimal Supersymmetric Standard Model a parity of exactly this kind is what forbids the proton from decaying through dimension-four operators.

Theorem 8.2 (matter parity is the Peirce symmetry)

COMPUTATION

For every one of the 27 coordinate idempotents c , the Peirce charge

$$Q = 6L_c - 2 \quad (\text{eigenvalue } 4 \text{ on } A_1, 1 \text{ on } A_{1/2}, -2 \text{ on } A_0)$$

reproduces the certified spectrum of Q_ψ , is conserved on all 45 triads, and gives exactly the pattern $40 \times (-2, 1, 1) + 5 \times (-2, -2, 4)$. Its parity $(-1)^Q$ is the Peirce symmetry U_{2c-e} , an automorphism of A that is +1 on $A_1 \oplus A_0$ and -1 on the spinor $A_{1/2}$. For a single-plane rotation generator $D \in \mathfrak{spin}(9)$, $\exp(2\pi D) = U_{2c-e}$ (numerical agreement 2.7×10^{-15}). w33_pass10950_clock_albert_lorentz_spinor.py

What it means physically

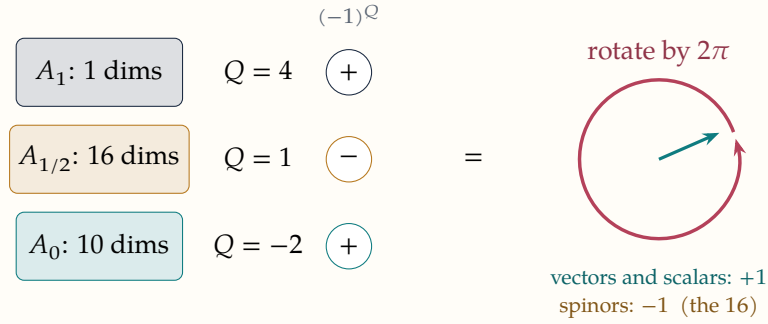


Figure 24. Matter parity in the clock Albert algebra: the sign that a full turn gives a spinor.

A rotation by 2π returns every vector to itself, but multiplies every spinor by -1 : the mathematical root of the difference between fermions and bosons. In $SO(10)$ grand unification it is known that matter parity can be realised by exactly such a central element of $\text{Spin}(10)$, acting as -1 on the matter **16** and $+1$ on the Higgs **10** (cf. [28]). Theorem 8.2 is the objectwise version of that fact inside the forty-point clock: the repository's matter parity *is* the 2π rotation of the $\text{Spin}(9)$ fixing a primitive idempotent. The same $27 = 1 + 10 + 16$ that a GUT reads as singlet, Higgs and matter is, in the clock real form, scalar, Lorentz vector and Weyl spinor. **BRIDGE** (reading) **COMPUTATION** (identity)

8.5 The clock's fourth tick is the same sign

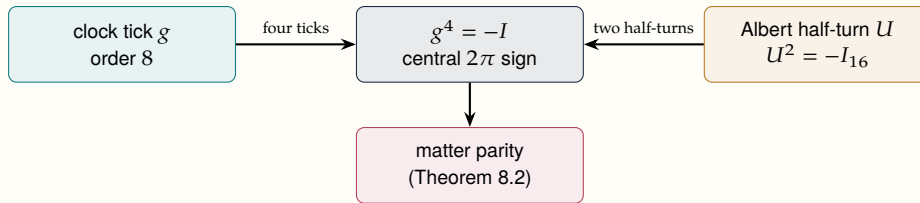
Section 4 met a clock whose fourth tick is a sign flip. The other research track has since shown that this is not an analogy but one spin structure, seen at both ends of the ladder. We cite their results (certificates named) rather than re-derive them.

Theorem 8.3 (clock, spin and matter parity; Passes 10951–10959)

COMPUTATION

- (i) The negacyclic clock tick lifts uniquely to $g = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \in GL(2, 3)$ of order eight, with $g^4 = -I$; under an explicit isomorphism $SL(2, 3) \cong 2T \subset \text{Spin}(3)$ the element $-I$ is the quaternion -1 , the central 2π spin sign. `w33_pass10951_clock_pin_spin_central_sign_bridge.py`
- (ii) As a signed permutation of the four clock coordinates, determinant-odd ticks exchange the two D_4 half-spin sheets and determinant-even ticks preserve them. `w33_pass10955_d4_halfspin_clock_bridge.py`
- (iii) On the Albert Peirce 16, the $\text{Spin}(8)$ fixing a spatial axis splits it as $8_s + 8_c$; an exact rational $\text{Spin}(9)$ half-turn U exchanges them, with $U^2 = -I_{16}$ — the matter-parity sign of Theorem 8.2 — and $U^4 = I_{16}$. `w33_pass10956_albert_spin8_halfspin_normalizer.py`, `w33_pass10958_exact_albert_mu4_missing_mode_intertwiner.py`
- (iv) One Albert 16 cannot carry the whole order-eight clock as a $GL(2, 3)$ representation; the minimal carrier is $16 \oplus 16$. `w33_pass10959_albert_clock_extension`

obstruction.py

**Where this stops**

The $\mathfrak{so}(1,9)$ here is the stabiliser of an idempotent inside $E_{6(-26)}$, an *internal* symmetry of the algebra; it is not derived to be the Lorentz group of physical space-time, and the 2π rotation is a rotation in that internal $\text{Spin}(9)$. The identification of Q_ψ with $6L_c - 2$ holds up to the $W(E_6)$ -transitive choice of coordinate idempotent. Neither chirality selection, masses nor gauge dynamics follow.

The exceptional ladder closes: from one qutrit, through $W(3,3)$, E_8 , a Hesse clock and its real form, to the Albert algebra $H_3(\mathbb{O})$ — whose automorphisms are F_4 , whose structure group is $E_{6(-26)}$, and inside which sit ten-dimensional Minkowski space, its Lorentz algebra $\mathfrak{so}(1,9)$, a Majorana–Weyl spinor, and, as a rotation by 2π , the matter parity that protects the proton.

9 The Standard Model on forty points

Sections 2–8 established exact structure. This section asks the question the programme exists to answer: how much of the actual physics of our universe does that structure carry? We give the answer at three levels — what the finite geometry itself contains, what a string compactification built on it produces, and what is still missing — and we end with a score-card.

9.1 What the finite geometry itself contains

The anatomy of one symmetry. Pick a point p_0 of $W(3,3)$ and the *transvection* R centred on it: the symplectic map $x \mapsto x + \langle x, v \rangle v$, of order three. It fixes the 13 points of the plane $p_0^\perp - p_0$ and its 12 neighbours — and permutes the other 27 points in nine orbits of three.

Proposition 9.1 (the transvection spine)

COMPUTATION

The centraliser $C(R)$ of a transvection has order 648 (it is the point stabiliser $3^{1+2}:2A_4$) and centre $\langle R \rangle \cong \mathbb{Z}_3$. Its permutation module on the 12 neighbours of p_0 decomposes as $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8}$. On the 27 far points, R has eigenspaces $9 \oplus 9 \oplus 9$. `bt886_standard_model_spine.py`

EXACT	Quantum measurement structure (qutrit Paulis, contexts, Clifford group)
EXACT	Exceptional algebras E_6, E_7, E_8, F_4 and the Albert algebra
EXACT	A finite light cone; clocks; one Lorentzian real form $\mathfrak{so}(1,9)$
HOSTED	Gauge group $SU(3) \times SU(2) \times U(1)$
HOSTED	Three generations
EXACT	One chiral generation as a spinor (16 of $\text{Spin}(1,9)$)
NO-GO	Which chirality: exchanged only by antiunitary (CPT-type) symmetries
EXCLUDED	Matter parity in a supersymmetric singlet vacuum (heterotic class)
NOT YET	A supersymmetric, realistic vacuum (nonabelian directions)
OPEN	Masses, mixing angles, coupling constants
OPEN	Gravity, a dynamical space-time, the cosmological constant

Figure 25. Scorecard. EXACT: a theorem or exact computation in this paper. HOSTED: realised in an explicit construction built on the geometry, not forced by it. EXCLUDED: exhaustively ruled out in a stated scope (Section 9.4). NO-GO: proved impossible from inside. OPEN: not addressed by any exact result.

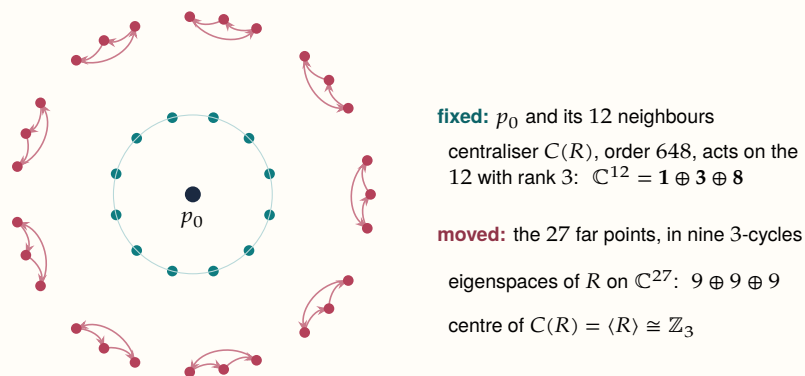


Figure 26. One transvection of $W(3,3)$. Its centraliser acts on the twelve fixed neighbours with the dimensions of the Standard Model's gauge algebra, $1 + 3 + 8$, and it splits the twenty-seven moved points into three equal eigenspaces.

How far this goes

The dimensions 1, 3, 8 are those of the Lie algebras $\mathfrak{u}(1)$, $\mathfrak{su}(2)$, $\mathfrak{su}(3)$, and the corpus has sometimes read Proposition 9.1 as “the Standard Model gauge group follows with zero free parameters”. That is an over-read. The $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{8}$ here are representations of a *finite* group of order 648; they are not Lie algebras, they do not come with gauge fields, and nothing fixes their couplings. The honest statement is that the local symmetry of one point of the geometry has the *shape* of the Standard Model’s gauge algebra.

BRIDGE

Three from the Steinberg module. The first homology of $W(3,3)$ (its 40 points, 240 edges and 160 triangles in lines) is 81-dimensional and irreducible under $\mathrm{PSp}(4,3)$: it is the *Steinberg module*, the most rigid representation a group of Lie type has, and the logical space of the repository’s $[[240, 81, 3]]_3$ quantum code. **COMPUTATION** (`bt861_code_register_is_steinberg.py`)

Theorem 9.2 (every order-three symmetry splits matter $27 + 27 + 27$)

THEOREM

Let $g \in \mathrm{PSp}(4,3)$ have order three. Then g acts on the 81-dimensional Steinberg module with eigenvalues $1, \omega, \omega^2$, each with multiplicity exactly 27.

Proof. The Steinberg character vanishes on every element whose order is divisible by the defining characteristic $p = 3$ (a theorem of Steinberg; see [18]). So $\chi(g) = \chi(g^2) = 0$, and the multiplicity of each eigenvalue ω^j is $\frac{1}{3}(\chi(1) + \omega^{-j}\chi(g) + \omega^{-2j}\chi(g^2)) = 81/3 = 27$. $\square$

Three is forced, but not selected

Theorem 9.2 holds for *every* element of order three, so it produces a threefold splitting without choosing one. And the string-theory track (below) found that in the explicit compactifications the number of families is the number of complex planes of the internal torus, not a property of $W(3,3)$. We therefore record the Steinberg splitting as a structural host for three generations, not a derivation of them. **BRIDGE**

The qutrit register is $SU(9)$, not E_6 . One might hope that the E_6 of grand unification acts directly on qutrit registers.

Theorem 9.3 (no Pauli E_6)

THEOREM

COMPUTATION

Conjugation by the three-qutrit Pauli group splits $\mathfrak{gl}(27)$ into 729 distinct one-dimensional character spaces, so every conjugation-stable subalgebra is spanned by Pauli operators. A simple Pauli-spanned Lie algebra has dimension $3^a - 3^b$, i.e. one of 8, 24, 72, 80, 216, 240, 648, 720, 728. Since $\dim \mathfrak{e}_6 = 78$ is not among them, E_6 cannot act on a qutrit register through Pauli-stable generators. `Holotrade 1cdelf1`

The natural gauge algebra of the qutrit register is instead $\mathfrak{sl}_9$ — the 80 of Section 5 — which is exactly what appears in the string construction.

9.2 The string that carries the geometry

The most developed attempt to turn this structure into physics uses the $E_8 \times E_8$ heterotic string compactified on an orbifold [8]: six extra dimensions curled up as three two-dimensional tori, divided by a finite rotation group. Its advantage is that everything is computable exactly, with a public tool, the ORBIFOLDER [29].

Theorem 9.4 ($W(3,3)$ lives in the twisted sector of E_8)

COMPUTATION

For the $\mathbb{Z}_3$ twist of the E_8 lattice vertex algebra, the commutator form of the twisted-sector Pauli group equals twice the lattice form on every rung, and the resulting commutation geometry is $W(3,3)$. The twist is anomalous on a single E_8 (ground-state energy $\rho = 4/9$), consistent exactly when $3 \mid k$ on E_8^k ; the E_8^3 orbifold is the Niemeier lattice vertex algebra $V_{A_8^3}$ with algebra $\mathfrak{sl}_9^3$. Holotrade 68df33f

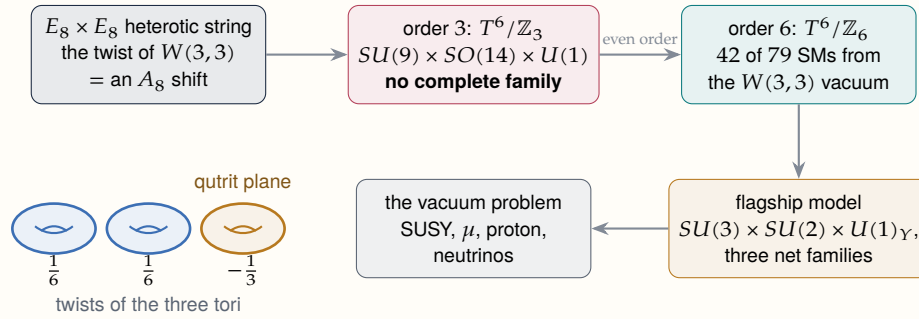


Figure 27. The heterotic route. At order three no Wilson-line breaking completes a Standard-Model family (a theorem); at even order the $W(3,3)$ vacuum is the most productive source of Standard Models; the open problem is a realistic vacuum. The third torus, the unique order-three plane, is where the Higgs lives.

Order three fails, for a reason. In the $T^6/\mathbb{Z}_3$ orbifold built on this twist, Wilson lines produce three-family $SU(5)$ and $SO(10)$ models; but an order-three Wilson line keeps at most one of Q, u^c, e^c in each $\mathbf{10}$, so no breaking completes a Standard-Model family. The general law: for a $\mathbb{Z}_N$ orbifold, u^c and e^c survive together only when N is even. This explains, from first principles, why heterotic model builders moved to even-order orbifolds. An exhaustive census of 127 980 classes confirms it. **THEOREM** **COMPUTATION** (Holotrade ed1eaca, a6e1c69, a779053)

Order six succeeds. Among the 58 $\mathbb{Z}_6$ -I and 61 $\mathbb{Z}_6$ -II shifts, the $W(3,3)$ vacuum $SU(9) \times SO(14) \times U(1)$ is the most productive class: of 79 inequivalent Standard Models found in it, 42 come from the $\mathbb{Z}_6$ -I $W(3,3)$ vacuum. The flagship has gauge group $SU(3)_C \times SU(2)_L \times U(1)_Y$ with three net families and no anomalies, and at one fixed point the local gauge group is $SU(9)$ with all four simple roots of the Standard Model's $SU(5)$ inside it: the Standard Model's unification group descends from the twist's own $\mathfrak{sl}_9$. **COMPUTATION** (Holotrade 5b3f3ad)

Measured regularities of the $W(3,3)$ Standard Models

COMPUTATION

- **The breaking is an entangling gate.** The order-three Wilson line’s $\mathfrak{sl}_3$ holonomy is of two-qutrit controlled-Z type $(5,2,2)$ in 45 of 45 $\mathbb{Z}_6$ -I models, against a 33% background (chance $\approx 2.6 \times 10^{-22}$). (d772153)
- **Two commuting holonomies cut out $SU(3) \times SU(2)$:** in all 25 models with parity-type θ^3 , against a 30% background ($\approx 8.6 \times 10^{-14}$). (94df376)
- **A heavy top is allowed:** 57 of 60 models admit a renormalisable top Yukawa coupling. (93b34e1)
- **The Higgs sits alone in the qutrit plane:** in all 87 $\mathbb{Z}_6$ -I models the weak doublets come only from the third torus (the unique order-three plane) and colour triplets only from the other two; this one fact gives both the top coupling and **doublet–triplet splitting**, solved in 55 of 87 models by a missing-partner mechanism. (3caf15e, a6f1cae)

What it means physically

The Standard Model’s most puzzling structural feature — that the Higgs doublet is light while its would-be colour-triplet partner, which would make protons decay, is absent — is here a consequence of geometry: the Higgs lives on the one internal plane whose twist has order three, the “qutrit plane”, and that plane carries no colour triplet. And the symmetry breaking that produces the Standard Model is, in qutrit language, an entangling gate. These are measured regularities of an explicit string construction, with controls; they are not theorems about all constructions.

9.3 Where the string route stands

A model is not a universe until it has a stable vacuum with the right low-energy properties. Here the honest record is mixed, and it has been corrected more than once.

Requirement	Current exact status	Badge
Supersymmetric vacuum	In the doublet–triplet-solving vacuum, F - and D -flatness conflict: one $U(1)$ direction gives every mass singlet a positive charge. Gordan’s alternative (“ D -flat or empty superpotential, never both”) verified on all 215 models.	REFUTED (that vacuum)
D -flat directions	Exist in 23 of 87 ($\mathbb{Z}_6$ -I) and 105 of 128 ($\mathbb{Z}_6$ -II); 83 of the 211 anomalous models have none.	COMPUTATION
Proton stability	No supersymmetric singlet vacuum preserves a matter parity: 0 of 88 models over the full $B - L$ freedom, 0 of 1024 $\mathbb{Z}_2$ parities (Section 9.4).	COMPUTATION (excluded)
The μ term	Arises at order 3 or 4 in every model; none is μ -free through order five. Must come from a vacuum symmetry.	OPEN
Neutrino masses	In the splitting-solving models, one to two orders of magnitude too light.	OPEN

9.4 The Fayet–Iliopoulos term forces a right-handed sneutrino

Every model of the class has an anomalous $U(1)$, whose Fayet–Iliopoulos term must be cancelled by giving vacuum values to some singlet fields (D -flatness), with $\langle n \rangle / M_s \approx 0.2$ – 0.4 . Proton stability needs a *matter parity* — a $\mathbb{Z}_2$ under which quarks and leptons are odd and Higgs fields even — to survive that condensation, so the fields that condense must all be even. The earlier record found every tested D -flat direction contained an odd singlet, but it had *sampled* 60 choices of $B - L$ per model and said explicitly that this was not a proof; the raw spectra needed to finish it were in no repository. They were recovered for this paper from the original scan and frozen as an exact-rational ledger of all 215 models.

Theorem 9.6 (no matter parity survives the FI term)

COMPUTATION

Over the *entire* rational solution set $B - L = x_0 + Nt$ of each model:

- (i) 64 of the 88 models with a $B - L$ direction have no D -flat singlet direction at all (an exact Farkas vector certifies that the FI term cannot be cancelled);
- (ii) in the other 23 $\mathbb{Z}_6$ -I models, the singlets that are even for *some* choice of $B - L$ are never D -flat (one exact Farkas vector each); the only singlets that are never even carry $3(B - L) = 3$, i.e. $B - L = +1$, fixed by the Standard Model itself, and every D -flat direction uses one;
- (iii) in the one remaining ($\mathbb{Z}_6$ -II) model, all 304 FI-cancelling extreme rays of the D -flat cone (3334 rays, computed exactly) are obstructed by an integer (Smith normal form) parity certificate; its never-even singlets carry $B - L = \pm 1$.

Independently, all 1024 $\mathbb{Z}_2$ characters of the charge lattice that are odd on every q, u^c, d^c, e^c give no even D -flat direction. Controls: the same exact machinery finds the $23 + 105 = 128$ D -flat models and the 88 $B - L$ models of the earlier record. `w33_pass10960_matter_even_dflat_closure.py`

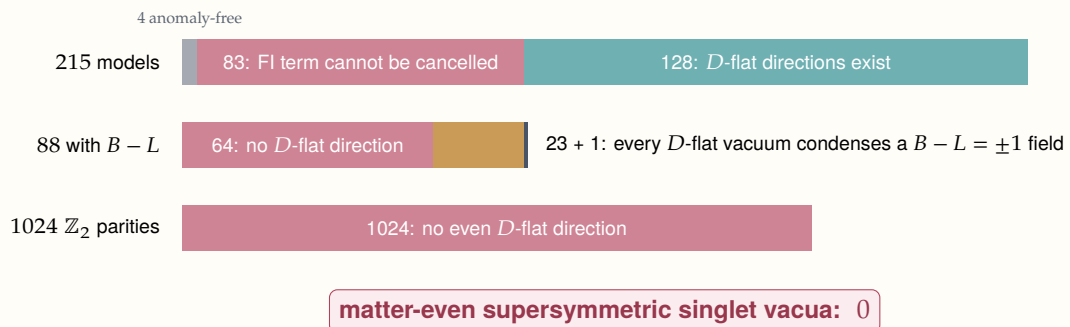


Figure 28. The exhaustive closure of Theorem 9.6. Every one of the 215 models is accounted for; no branch leaves a matter-parity-preserving supersymmetric vacuum in singlet directions.

What it means physically: the sneutrino that breaks R-parity

A singlet with $B - L = +1$ has the quantum numbers of a right-handed (s)neutrino. Theorem 9.6 says the anomalous $U(1)$ *forces* such a field to condense. Martin’s criterion [28] is exactly relevant here: matter parity survives the breaking of $B - L$ only if the fields that break it have even $3(B - L)$ — which is why realistic $SO(10)$ models break

$B - L$ with **126**-type fields ($B - L = \pm 2$) rather than **16**-type ones ($B - L = \pm 1$). In this class the FI term leaves no choice: $B - L$ is broken by one unit, and the baryon- and lepton-number-violating couplings that matter parity was protecting are no longer forbidden by any symmetry, at the scale of the singlet vacuum values.

The hidden sector does not rescue it. Theorem 9.6 covers singlets of every nonabelian group. Fields that are Standard-Model singlets but charged under the *hidden* gauge group can also condense, and their parity can be compensated by a hidden gauge element. Pass 10962 attacks this loophole in tiers:

- **exact, for every hidden direction:** a parity-preserving vacuum must be D -flat for the hidden maximal torus with component-wise parity (up to a hidden torus element); for 32 of the 88 models even this relaxation has an exact Farkas certificate;
- **exact over explicit invariant generators:** by the Luty–Taylor and Kempf–Ness correspondence a parity-preserving D -flat vacuum exists iff an FI-cancelling product of matter-even hidden-group invariants does; over singlets, all quadratic invariants, $SU(N)$ baryons and $SU(5)$ cubics, the other 55 models close by exact Farkas vectors, and the last by a parity obstruction;
- the four anomaly-free models (no FI term) admit no matter parity at all.

COMPUTATION (`w33_pass10962_hidden_sector_dflat_tiers.py`)

Scope

For the 56 models not closed at the first tier, completeness of the invariant generators is not proved (they contain bi-fundamental, $SU(4)$ **6** or $SU(5)$ **10** hidden fields), so for them the hidden-sector statement is “no rescue among the standard invariants” rather than a theorem. Whether higher-order selection rules suppress the regenerated couplings in a particular model is not addressed. **OPEN**

9.5 Handedness: host, not select

The weak force acts only on left-handed particles. Can $W(3,3)$ explain why?

Theorem 9.5 (the chirality no-go)

THEOREM

COMPUTATION

The two half-spin representations $S^\pm$ of the D_5 structure built on the geometry are exchanged by the substrate’s own outer controller T , which generates $\mathrm{PGSp}(4,3) = \mathrm{PSp}(4,3):2 = W(E_6)$ together with $\mathrm{PSp}(4,3)$ and has $\det T = -1$. Consequently no invariant of the full automorphism group of $W(3,3)$ can distinguish S^+ from S^- . `w33_pass346_the_chirality_no_go.py`

In plain words

The forty-point geometry is perfectly capable of *carrying* a left-handed world: the chiral spinors are there (Section 8). What it cannot do is *choose* left over right, because

one of its own symmetries swaps them. Any explanation of the handedness of the weak force must come from outside the geometry — from a choice of vacuum, of orientation, or of dynamics. This is the same distinction as in Section 2: the two groups of order 51 840 differ precisely by whether that swapping symmetry is present.

What it means physically: the swap is antiunitary

Which symmetries do the swapping? For the qutrit, the determinant of a symplectic label map decides: determinant +1 is a unitary Clifford operation, determinant -1 can only be implemented antiunitarily (Pass 5730, `w33_pass5727_5730_torsion_family_heisenberg.py`). The decisive statement was proved on the 32 orientations of a complete two-qutrit factorisation frame (Holotrade `e92a047`, `frame_chirality_is_reversed_only_by_antiunitaries.py`): they split $16+16$ exactly as the two half-spin weight sets of D_5 , unitary Cliffords preserve each half, and antiunitary ones — and nothing else — exchange them. The other track has since found the same law on the clock: determinant-odd ticks exchange the two D_4 half-spin sheets (Pass 10955) and are not completely positive (Pass 10952). So the geometry's chirality is invariant under every *unitary* symmetry and exchanged only by *antiunitary* ones. That is the structure of a chiral quantum theory obeying CPT: the mirror of a chiral world is related to it by an antiunitary map, and which of the two is called “left” is a convention that no internal measurement can fix. The no-go is the finite form of this, not a defect peculiar to $W(3,3)$. **BRIDGE** (`w33_pass10952_clock_complete_positivity_firewall.py`, `w33_pass10955_d4_halfspin_clock_bridge.py`)

Corollary 9.7 (one coset reverses time, acts antiunitarily, and swaps chirality)

THEOREM

Fix a “now” (the Bell line B). Its stabiliser in $\mathrm{PGSp}(4,3)$ has order 1296, and its elements outside $\mathrm{PSp}(4,3)$ form a single coset of size 648. That coset is exactly

- (i) the set of symmetries that *reverse* the unique global orientation of the history graph (Theorem 4.4);
- (ii) the set that can only be implemented *antiunitarily* on qutrits (Pass 5730);
- (iii) the set that *exchange* the two chiral half-spin modules (Theorem 9.5; Holotrade `e92a047`).

Proof. Each item identifies its set with the complement of $\mathrm{PSp}(4,3)$ in the relevant group: (i) is the statement of Theorem 4.4(ii) (the 648 elements of the Bell stabiliser in $\mathrm{PSp}(4,3)$ fix the orientation cycle, the other 648 negate it); (ii) symplectic similitudes of multiplier +1 lift to unitary Clifford operations and those of multiplier -1 only to antiunitary ones; (iii) $\mathrm{PSp}(4,3)$ preserves S^+ and S^- (they are inequivalent) while the outer element T exchanges them. Intersecting with the stabiliser of B gives the same coset in all three cases. $\square$

What it means physically: a finite CPT theorem

In quantum field theory, the CPT theorem says that the only way to relate a chiral world to its mirror image by a symmetry is antiunitary, and that the same operation

reverses the direction of time. Corollary 9.7 is that structure, exactly, in the forty-point geometry: the operations that reverse time's orientation are the antiunitary ones, and they are precisely the ones that exchange left and right. Choosing an orientation of time does not choose a chirality — but it makes chirality a conserved label, and the world of opposite chirality is the time-reversed one. This answers, in the finite model, the question of whether the datum that fixes the arrow and the datum that fixes handedness are related: they live in the same coset. **BRIDGE**

9.6 What we do not count as evidence

The older manuscripts contain many numerical coincidences: $1/\alpha \approx 137$ written as a combination of cyclotomic values of $q = 3$, spectral indices of the cosmic microwave background, spanning-tree counts read as gravitational entropies. Some are striking. None is derived from a dynamical principle with its uncertainties, and a finite geometry with many natural integers can match many targets. We therefore do not use any of them as evidence in this paper. (The value $\sin^2 \theta_W = 3/8$ at unification, often quoted alongside them, is the standard prediction of any $SU(5)$ or $SO(10)$ grand unification and is not specific to $W(3,3)$.) **BRIDGE** at most.

The geometry carries the Standard Model's shapes — its gauge algebra's dimensions, three equal matter sectors, a chiral spinor, a matter parity — and a string compactification built on its twist produces genuine three-family Standard Models whose Higgs sits in the "qutrit plane". It cannot choose the handedness of the weak force — only an antiunitary, CPT-type symmetry exchanges the two — and, exhaustively, none of its supersymmetric singlet vacua preserves a matter parity: the anomalous $U(1)$ forces a $B - L = \pm 1$ condensate.

10 The machine: a computer read off the geometry

The same forty points that carry E_8 and a light cone also describe, with no design choices, a quantum computer: an instruction set, a memory, a network and a supply of the non-classical resource that makes quantum computation powerful. The Holonet papers develop this at length; here we give the parts that are exact, and mark carefully what has and has not been built.

10.1 The carrier: two qutrits in one photon

A single photon has several independent degrees of freedom. Two of them — which of three *time bins* it arrives in, and which of three *frequency bins* it occupies — give two qutrits, and together the $\mathbb{C}^9$ on which the two-qutrit Paulis of Section 2 act. Its path and polarisation give a second register, $\mathbb{C}^4$, whose 40 Witting rays realise $W(3,3)$ as *states* (Theorem 5.3 and the box after it).

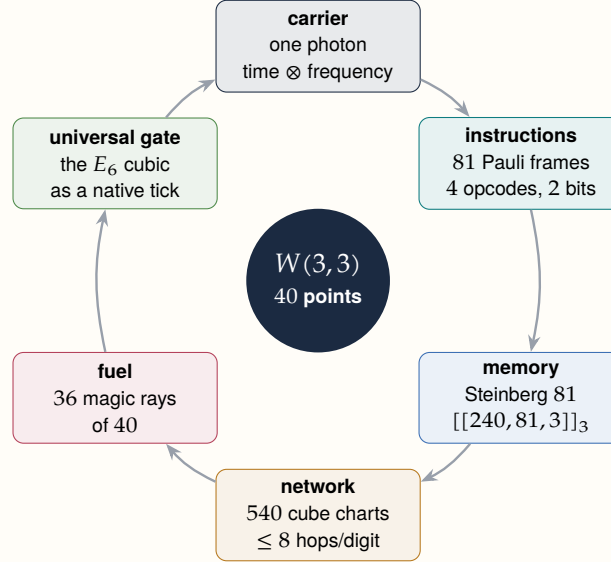


Figure 29. The Holonet stack. Every layer is a structure of $W(3,3)$ read as engineering: the Pauli frames are the 81 points of $\mathbb{F}_3^4$, the memory is the Steinberg module, the network is the geometry's atlas of cube charts, the fuel is its set of magic states, and the universal gate is the E_6 cubic of Section 5.

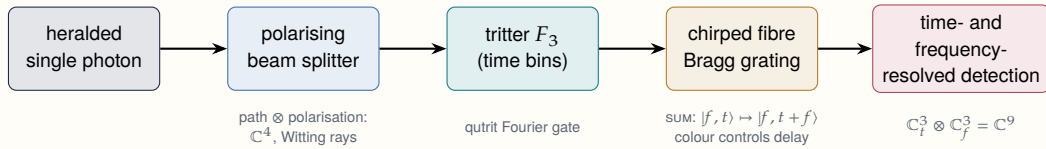


Figure 30. One photon, two qutrits. The controlled-add (sum) gate between the frequency and time qutrits has been demonstrated with a single passive element, a chirped fibre Bragg grating, at basis fidelity 0.92 ± 0.01 [19].

10.2 The instruction set

The machine’s classical control state is a *Pauli frame*: a label $x \in \mathbb{F}_3^4$ saying which Pauli correction is pending. There are 81 frames. Instructions are symmetries of the geometry: the two Fourier gates, the two phase gates, the two controlled-add gates, and the Pauli translations.

Theorem 10.1 (the complete instruction group)

COMPUTATION

The six Clifford opcodes generate exactly $\text{Sp}(4, 3)$ (order 51 840); adjoining one translation generates the affine group

$$\text{ASp}(4, 3) = \mathbb{F}_3^4 \rtimes \text{Sp}(4, 3), \quad |\text{ASp}(4, 3)| = 81 \times 51\,840 = 4\,199\,040.$$

No one or two linear opcodes suffice; exactly six triples do, so *four* instructions (three Clifford, one translation) in a *two-bit* opcode field generate everything, with worst-case program length 19 for the best triple. The minimal arithmetic core synthesises to 43 logic cells at 208.86 MHz on an iCE40 HX8K FPGA. `holonet_machine_blueprint_body.tex`, Passes 2778, 2789, 2882

In plain words

Four instructions are enough to perform every one of more than four million possible frame transformations, and no program needs more than nineteen steps. The instruction set was not designed: it is the list of symmetries of the forty-point shape, and the theorem says which symmetries you can leave out.

10.3 The fuel: matter is magic

Clifford operations alone can be simulated efficiently on an ordinary computer; quantum advantage needs a non-Clifford resource, usually supplied as *magic states*; contextuality is what supplies them [17].

Theorem 10.2 (magic is everything except one line)

COMPUTATION

Of the 40 Witting rays, exactly the 4 computational-basis rays are stabiliser states; the other 36 are magic. The 36 split by stabiliser fidelity into grades $8 + 24 + 4$. A classical $\{0, 1\}$ assignment can satisfy the “exactly one” rule on at most 36 of the 40 contexts, so the contextual fraction is exactly $1/10$; the independence number of the collinearity graph is 7, attained by heptads of seven rays with all pairwise overlaps $1/3$. `bt822_magic_census_matter_is_magic.py`, `bt818_ovoid_nogo_theta_gap.py`

The cubic gate. Clifford gates are “degree two”: they act on frame labels by linear symplectic maps and on phases by quadratic forms. The classical continuous-variable theorem of Lloyd and Braunstein [26] says that quadratic operations plus a single cubic one are universal. In Section 5 the cubic appeared as the E_6 form N , the rule by which three first-order moves reach the top of the ladder. Pass 10947 compiles that signed cubic into a native gate on a seven-qutrit virtual machine: ten operations in seven parallel layers, both ancillas re-

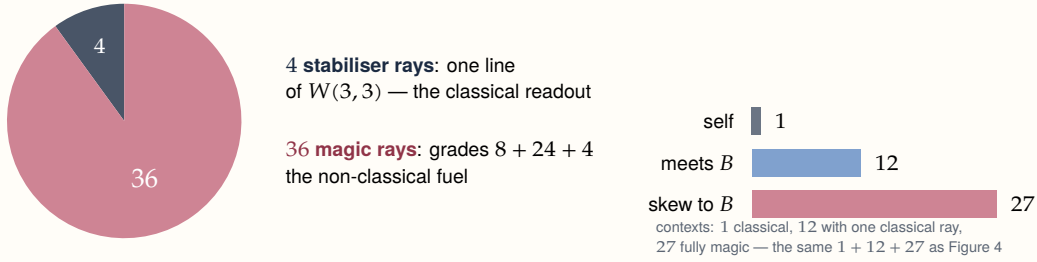


Figure 31. The machine’s quantum resource is the geometry itself: remove one line and every remaining point is a magic state.

turned to zero. **COMPUTATION** (`w33_pass10947_five_front_execution.py`) The same object is, in the language of Section 5, the Yukawa coupling of E_6 unification, and in the language of this section, the machine’s universal non-Clifford gate. **BRIDGE**

10.4 Memory and error correction

Theorem 10.3 (the memory is the Steinberg module)

COMPUTATION

The CSS code $[[240, 81, 3]]_3$ on the edges of $W(3,3)$, with distances $(d_X, d_Z) = (3, 4)$, stores its 81 logical qutrits in $H_1(W(3,3))$, which is the irreducible Steinberg module of $\text{PSp}(4, 3)$. By Schur’s lemma any symmetry-covariant logical error acts as a scalar. Separately, a ternary Golay $[[11, 1, 5]]_3$ lane with two verified syndrome rounds has 552 fault locations in a declared model, 2 576 205 malignant fault triples, and a conservative threshold crossing at $p \approx 5.32 \times 10^{-4}$. `bt861_code_register_is_steinberg.py`, `w33_pass10947_five_front_execution.py`

10.5 The network

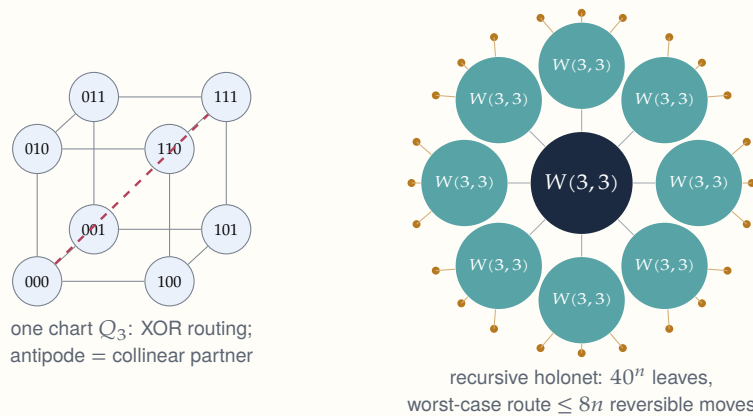


Figure 32. Left: a local routing chart. Each of the 540 pairs of disjoint lines of $W(3,3)$ carries a cube of eight points with XOR addressing. Right: substituting a copy of $W(3,3)$ at every point gives a network with 40^n leaves and logarithmic routing.

$W(3,3)$ contains exactly 540 pairs of disjoint lines; each pair’s eight points carry the cube

graph Q_3 , in which the antipode of a vertex is its unique collinear partner. The charts are glued along the 1620 apartments of the Tits building of $\mathrm{Sp}(4,3)$. A route needs at most 3 in-chart moves plus 5 chart hops, so each digit of a recursive address costs at most 8 reversible moves. **COMPUTATION** (bt773_involution_cube_theorem.py, bt777_hypercube_chart_web.py, bt827_holonet_fractal_architecture.py)

What is measured, what is proved, what is not built

Measured in a laboratory (prior art): the single-photon time–frequency SUM gate, fidelity 0.92 ± 0.01 [19]. **Proved exactly:** the group theory, the instruction semantics, reachability, code parameters, routing bounds, and FPGA synthesis figures. **Not built:** an integrated device; a magic-state injection circuit with a threshold; a physical demonstration of the network. The machine is a specification whose every number can be checked, not a device.

The forty points are also a computer: 81 frames, four instructions generating 4 199 040 operations, a memory that is the Steinberg module, a network of 540 cube charts, and a quantum fuel that is every point off one line. The E_6 cubic that is the Yukawa coupling in physics is the universal gate in the machine.

11 Epilogue: what a theory of everything still needs

We began with a qutrit acting on itself and followed one chain of exact identifications: forty points; a finite present with a light cone; four clocks; E_8 from nine histories; the Freudenthal quartic and a 57-dimensional light cone; a real form in which time has one dimension; the Albert algebra, the Lorentz algebra $\mathfrak{so}(1,9)$, a chiral spinor, and matter parity as a 2π rotation. On top of that chain a heterotic string construction produces three-family Standard Models, and the same geometry specifies a quantum computer.

exact	bridge / hosted	open
• $W(3,3)$ as qutrit commutation geometry • Bell line = unitary channels • 27-event light cone, $t^2 - x^2 - y^2$ • tetracode = four clocks • $E_8 = 80+84+84$; $240 = 40 \times 6$ • clock gradings; cubic ladder • Freudenthal quartic; 57-cone • clock real form $\Rightarrow (1,9)$ • F_4, $E_{6(-26)}$, $\mathfrak{so}(1,9)$, 16 • matter parity = 2π rotation • chirality no-go (antiunitary swap) • no matter parity in SUSY singlet vacua (all 215 models) • ISA, memory, magic census	• “null” = shared measurement • lifted history = proper time • forgetting 2 trits = arrow • cubic ladder = cubic time • quartic = black-hole entropy • $1+3+8$ = gauge shape • $27+27+27$ = generations • $W(3,3)$ twist $\Rightarrow$ heterotic SMs • Higgs in the qutrit plane • SM breaking = CZ gate	• a dynamical principle • a continuum limit; gravity • which chirality, and why • SUSY vacuum via hidden nonabelian condensates • masses, mixings, couplings • the 54 Fourier intertwiner • magic-state injection • a built device

Figure 33. The state of the programme, in three columns.

11.1 Seven open problems, stated sharply

1. **A realistic vacuum.** In singlet directions this is now settled negatively (Theorem 9.6): the FI term forces a $B - L = \pm 1$ condensate in every model. What remains is the hidden nonabelian sector: is there a D -flat direction built from hidden $SO(10)$ or $SU(2)$ condensates whose parity is compensated by a hidden centre element — or must the programme move to other orbifold groups ($\mathbb{Z}_{12}$, $\mathbb{Z}_2 \times \mathbb{Z}_4$, ...) of the same twist? Both are finite computations on the frozen ledger.
2. **A dynamical principle.** Every result in this paper is kinematics. What action, Hamiltonian or spectral principle on $W(3, 3)$ has the structures found here as its symmetries, and what does it predict? In particular: does such a principle select the clock real form $E_{8(-24)}$ over the split form — does the geometry *choose* one time?
3. **A continuum limit.** The finite light cones (in $2 + 1$, $3 + 1$, $1 + 9$ and the quartic 57) are exact algebraic objects. Is there a limit, a tower over $W(3, q)$ or a lattice construction, in which one of them becomes a smooth Lorentzian manifold with dynamical curvature?
4. **Chirality from outside.** Corollary 9.7 shows that reversing time's orientation, acting antiunitarily and exchanging chirality are one coset, so fixing an arrow makes chirality conserved but does not choose it. What physical datum correlates the two — which chirality goes with which direction of time — and can it come from a vacuum?
5. **The 54 intertwiner.** The diagonal clock/phase weld projects onto 54 Fourier directions, while the clock grading supplies a canonical $27 \otimes 2$ at degree one. Is there an explicit equivariant isomorphism?
6. **Numbers.** The up-type Yukawa couplings of the $\mathbb{Z}_6$ -I class have a $2 + 1$ block structure qualitatively like the observed quark mixing. Can any mass ratio or mixing angle be computed, with an error, from a vacuum of the class?
7. **The machine.** Specify a magic-state injection circuit for the 36 Witting rays with a decoder and a threshold, and build the two-qutrit photonic core.

11.2 Corrections we kept on the page

A research record is trustworthy only if it keeps its errors. These were believed, written down, and then shown false; each correction made the structure *more* natural, not less.

Was believed	Is true
there is a natural bijection $240 \text{ edges} \leftrightarrow 240 \text{ roots}$	no equivariant bijection; a 6:1 fibration of roots over <i>points</i> (§5.4)
the Freudenthal 56 is the clock's 54 plus 2	the 56 contains only one 27 of the 54 (§6.1)
the Albert algebra has “27 minimal idempotents” as frames	the frames are the 45 triads (§6.4)
three time bins hold a past and a future qutrit	$\mathbb{C}^3$ cannot hold two qutrits; use time $\otimes$ frequency (§3.4)
the 9 fixed modes of the spectral clock are the qutrit's 9 histories	same count, different module (§4.4)
the independence number of $W(3,3)$ is 10 (an ovoid)	it is 7; $W(3,3)$ has no ovoid (§10.3)
at most 34 of 40 contexts are classically satisfiable	exactly 36 (§10.3)
the $\mathbb{Z}_6$ -I class is excluded unconditionally	excluded only if matter parity is broken (§9.3)
every D -flat support contains a matter-odd singlet (from 60 samples)	exhaustively true, and the odd singlet always has $B - L = \pm 1$ (§9.4)
the three families come from $W(3,3)$	in the orbifolds they come from the three tori (§9.1)

The forty points do not yet give a theory of everything. They give something rarer: a single finite object in which quantum measurement, a light cone, clocks, E_8 , the octonionic Albert algebra, ten-dimensional Lorentz symmetry, a chiral spinor, matter parity and a quantum computer are exact aspects of one structure — with every claim checkable, and the remaining distance to physics mapped rather than hidden.

A The ledger: every claim, its status, its witness

Every script named below lives in `analysis/` of the repository `wilcompute/W33-Theory` and writes its certificate to `data/`; most have a regression test in `tests/`. Commit hashes marked “Holotrade” refer to the second research track and are recorded row by row in the ledger of the $W(3,3)$ paper.

Claim	Badge	Witness
<i>The object and one qutrit</i> (§§2–3)		
$W(3,3)$: 40 points, 40 lines, $\text{SRG}(40, 12, 2, 4)$, spectrum $12, 2^{24}, (-4)^{15}$	THEOREM	Props. 2.1–2.2 (proofs in text)
two-sided Weyl maps span $\mathfrak{sl}_9$; commutation geometry is $W(3,3)$	THEOREM COMPUTATION	w33_20260924_temporal_su9_e8_compiler.py
Bell line = unitary-channel diagonal; swap is anti-symplectic	THEOREM COMPUTATION	w33_20260924_liouville_swap_cp_outer_bridge.py

Claim	Badge	Witness
transpose induces the outer class (up to labelling gauge)	COMPUTATION	w33_20260924_intrinsic_transpose_outer.py
memory witness: Bell fidelity $1 - 8p/9$, threshold $p = 3/4$	COMPUTATION	w33_20260924_temporal_quantum_memory_witness.py
modular flow of a qutrit state has A_2 frequency pattern	COMPUTATION	w33_20260924_modular_a2_event_clock.py
point/line firewall (27/108/36 coincidences, non-isomorphic)	COMPUTATION	w33_20260924_history_pointline_cospectral_firewall.py
<i>Time</i> (§4)		
$40 = 1 + 12 + 27$; big cell of $Q(4, 3)$	COMPUTATION	w33_20260924_history_bigcell_q43_compactification.py
finite light cone $t^2 - x^2 - y^2$; $1 + 8 + 6 + 12$	THEOREM COMPUTATION	same
$\text{Sym}_2(\mathbb{F}_3) = \mathbb{F}_3^4 / \langle 1111 \rangle$; $D_0 + D_\infty + D_+ + D_- = 3I$	THEOREM COMPUTATION	w33_20260924_positive_null_counter_cover.py
null graph spectra; $U^3 = I$ at $t^* = 2\pi/9$	COMPUTATION	w33_20260924_null_history_spectral_clock.py
fixed 9 is not the qutrit module	REFUTED	w33_20260924_spectral_clock_fixed9_group_probe.py
36 triangles; unique orientation cycle; $82 = 1 + 81, 81 = 35 + 46$	COMPUTATION	w33_20260924_history_invariant_cycle_orientation.py
filled temporal H_1 injects into $H_1(W(3, 3))$	COMPUTATION	w33_20260924_history_to_global_h1_intertwiner.py
universal cover: 8-regular tree, $\pi_1 = F_{82}$	THEOREM COMPUTATION	w33_20260924_temporal_universal_cover.py
Weil phase census $1^{13}i^4(-1)^6(-i)^4$; $(FG)^3 = iF^2$	COMPUTATION	w33_20260924_temporal_maslov_mu12.py
common S_4 of null directions, Hesse clocks, A_2 factors	COMPUTATION	w33_20260924_null_hesse_4a2_s4_intertwiner.py
tetracode = clock evaluation code	COMPUTATION	w33_pass10946_clock_code_cone_objectwise.py
negacyclic hyperbolic pair, $N^4 = -I$ (other track)	COMPUTATION	w33_pass10948_clock_tetracode_negacyclic_hyperbolic_bridge.py
Page–Wootters diagonal weld; 24 vs 248	COMPUTATION	w33_20260924_page_wootters_diagonal_weld.py
Hermitian $3 + 1$: $21 + 30 + 30$; null directions = spread	COMPUTATION	w33_20260924_hermitian_3p1_spread_bridge.py
no compatible complex structure J	REFUTED	same

Claim	Badge	Witness
forgetting two trits gives the depolarising channel E_8 (§5)	THEOREM	Thm. 4.7 (proof in text); bt823_the_closure.py
$E_8 = \mathfrak{sl}_9 + \Lambda^3 + \Lambda^6$ as packet operations	COMPUTATION	w33_20260924_temporal_a8_e8_process_bracket.py
Hesse $4A_2$; $240 = 72 + 24 + 144$	THEOREM COMPUTATION	w33_20260924_temporal_hesse_4a2_a8_bridge.py
no equivariant edge–root bijection	COMPUTATION	w33_pass1020_e8_transitive_51840.g
$240 = 40 \times 6$ fibration onto points	COMPUTATION	w33_pass1021_e8_fibration_over_forty.g
Hesse clock gradings $1+56+134+56+1$ and $ 3 $	COMPUTATION	w33_20260925_hesse_clock_e8_gradings.py
$\mathbb{Z}_3$ grading is the mod-3 shadow	COMPUTATION	w33_20260925_e8_parabolic_cubic_clock_lift.py
cubic ladder $54 \rightarrow 81 \rightarrow 83$; 45 triads $\times 12$	COMPUTATION	w33_20260925_cubic_clock_bracket_ladder.py
<i>Freudenthal, real forms, Albert</i> (§§6–8) 56 shares one 27 with the 54; $54 + 2 = 56$ refuted	COMPUTATION REFUTED	w33_pass10949_freudenthal_quasiconformal_clock_cone.py
Freudenthal normal form of the quartic, no sign repair	COMPUTATION	same
57-cone, Heisenberg law, GKN inversion	COMPUTATION	same
$K(e_\alpha, e_{-\alpha}) = -60(-1)^{\text{ht } \alpha}$; split = height twist	COMPUTATION	same
tick lines: $E_{8(-24)}$, Klein four-group, $80 + 3 \times 56$	COMPUTATION	same
Euclidean Albert algebra, Peirce $1 + 16 + 10$, $(1, 9)$ vs $(5, 5)$	COMPUTATION	same
$\text{Der} = F_{4(-52)}$, $\text{str}_0 = E_{6(-26)}$	COMPUTATION	w33_pass10950_clock_albert_lorentz_spinor.py
$\mathfrak{so}(1, 9)$, Majorana–Weyl 16; $16 = (2, 4) + \overline{(2, 4)}$	COMPUTATION	same
$Q_\psi = 6L_c - 2$; matter parity $= \exp(2\pi D)$	COMPUTATION	same
clock $g^4 = -I$ is the central 2π spin sign (other track)	COMPUTATION	w33_pass10951_clock_pin_spin_central_sign_bridge.py
Albert $16 = 8_s + 8_c$; half-turn $U^2 = -I_{16}$; minimal full-clock carrier $16 \oplus 16$ (other track)	COMPUTATION	w33_pass10956_albert_spin8_halfspin_normalizer.py, w33_pass10959_albert_clock_extension_obstruction.py

Claim	Badge	Witness
<i>Physics (§9)</i>		
transvection spine: $648, 1 \oplus 3 \oplus 8, 9 \oplus 9 \oplus 9$	COMPUTATION BRIDGE	bt886_standard_model_spine.py
Steinberg: every order-3 element splits $27 + 27 + 27$	THEOREM	Thm. 9.2 (proof in text)
no Pauli-spanned E_6 on qutrit registers	THEOREM COMPUTATION	Holotrade 1cde1f1
$W(3, 3) =$ Pauli geometry of $\mathbb{Z}_3$ -twisted V_{E_8}	COMPUTATION	Holotrade 68df33f
order-3 orbifold completes no SM family	THEOREM COMPUTATION	Holotrade ed1eaca, a6e1c69, a779053
$\mathbb{Z}_6$: 42 of 79 SMs from the $W(3, 3)$ vacuum	COMPUTATION	Holotrade 5b3f3ad
CZ-type breaking 45/45; $SU(3) \times SU(2)$ cut 25/25	COMPUTATION	Holotrade d772153, 94df376
Higgs alone in the qutrit plane; DTS 55/87	COMPUTATION	Holotrade 3caf15e, a6f1cae
F/D conflict; Gordan alternative 215/215	COMPUTATION	Holotrade babfd48
matter parity in all 87; class closure conditional	COMPUTATION	ledger commit 78ec0a0fc
every sampled D -flat support has a matter-odd singlet	COMPUTATION OPEN	ledger commit a3dbdf2d7
chirality no-go	THEOREM COMPUTATION	w33_pass346_the_chirality_no_go.py
chirality swap is exactly the antiunitary (determinant-odd) coset	COMPUTATION BRIDGE	Holotrade e92a047; w33_pass10952_clock_complete_positivity_firewall.py, w33_pass10955_d4_halfspin_clock_bridge.py
no matter-even SUSY singlet vacuum: 0/88 over full $B - L$, 0/1024 $\mathbb{Z}_2$ parities; forced $B - L = \pm 1$ condensate	COMPUTATION	w33_pass10960_matter_even_dflat_closure.py
hidden sector: 32 exact (all directions), 55 over invariant generators, 1 by 62,309 rays; anomaly-free models have no parity	COMPUTATION	w33_pass10962_hidden_sector_dflat_tiers.py, w33_pass10962_tierc_rays.py
time reversal = antiunitary = chirality swap (one coset)	THEOREM	Corollary 9.7 (proof in text)
<i>Machine (§10)</i>		
$\text{ASp}(4, 3)$, 4 199 040; four opcodes; length 19	COMPUTATION	blueprint, Passes 2778/2789/2882
4 stabiliser + 36 magic rays; KS budget 36/40; $\alpha = 7$	COMPUTATION	bt822_magic_census_matter_is_magic.py, bt818_ovoid_nogo_theta_gap.py

Claim	Badge	Witness
Steinberg memory $[[240, 81, 3]]_3$	COMPUTATION	bt861_code_register_is_steinberg.py
Golay lane: 552 locations, threshold 5.32×10^{-4}	COMPUTATION	w33_pass10947_five_front_execution.py
cube-chart atlas; ≤ 8 moves per digit	COMPUTATION	bt777_hypercube_chart_web.py, bt827_holonet_fractal_architecture.py
single-photon SUM gate, 0.92 ± 0.01	measured	[19]

B Glossary

A_2 The root system of $SU(3)$: six vectors forming a regular hexagon.

Albert algebra The 27-dimensional algebra $H_3(\mathbb{O})$ of 3×3 Hermitian octonionic matrices; the only exceptional Euclidean Jordan algebra.

Bell line The line of $W(3, 3)$ formed by the unitary channels $A \mapsto P_u A P_u^\dagger$; the “now” of Section 4.

Bridge A true mathematical statement together with a proposed physical reading.

Cartan involution An involution of a real Lie algebra splitting it into rotations ($\mathfrak{k}$) and boosts ($\mathfrak{p}$); it fixes the real form.

Clifford group Quantum operations mapping Pauli operators to Pauli operators; for two qutrits it acts on labels as $\text{Sp}(4, 3)$.

Context A maximal set of simultaneously measurable Pauli operators; a line of $W(3, 3)$.

Contextuality The impossibility of assigning pre-existing outcomes to all measurements consistently across contexts; a resource for quantum computation.

CSS code A quantum error-correcting code built from two classical codes; $[[n, k, d]]_q$ means n physical qudits, k logical, distance d .

E_8 The largest exceptional simple Lie algebra: dimension 248, 240 roots.

$\mathbb{F}_3$ The field $\{0, 1, 2\}$ with arithmetic modulo 3.

Freudenthal triple system A 56-dimensional space with a symplectic form and an E_7 -invariant quartic; the charge space of certain supergravity black holes.

Generalized quadrangle A point–line geometry with no triangles in which a point off a line is collinear with exactly one of its points.

Grading A splitting of an algebra into pieces labelled by integers (or integers mod n) that add under the bracket.

Hesse clock One of the four parallel classes of lines of $AG(2, 3)$, read as three ticks.

Heisenberg algebra The algebra of positions, momenta and a central element; here $56 + 1 = 57$ dimensions.

Heterotic string A closed string theory with gauge group $E_8 \times E_8$ (or $SO(32)$); on orbifolds it yields Standard-Model-like theories.

Lagrangian A maximal subspace on which the symplectic form vanishes; a line of $W(3, 3)$.

Magic state A state that is not a stabiliser state; injecting it makes Clifford circuits universal.

Majorana–Weyl spinor A real, chiral spinor; in ten dimensions it has 16 components.

Matter parity A $\mathbb{Z}_2$ symmetry, -1 on matter fermions and $+1$ on Higgs fields, that forbids fast proton decay in supersymmetric theories.

Octonions $\mathbb{O}$ The eight-dimensional, non-associative normed division algebra.

Orbifold A space divided by a finite group; used to curl up extra dimensions.

Pauli operator For a qutrit, $X^a Z^b$; for two qutrits, a tensor product of two.

Peirce decomposition The splitting of a Jordan algebra by the eigenvalues $1, \frac{1}{2}, 0$ of multiplication by an idempotent.

Qutrit A quantum system with three perfectly distinguishable states.

Real form A real Lie algebra whose complexification is a given complex one; E_8 has three.

Root An eigen-direction of the maximal commuting part of a Lie algebra.

Signature For a quadratic form, the numbers of positive and negative directions; for a real Lie algebra, $\dim \mathfrak{p} - \dim \mathfrak{k}$.

Spread A set of pairwise disjoint lines covering every point; $W(3, 3)$ has 36.

Steinberg module A distinguished irreducible representation of a finite group of Lie type; for $\mathrm{PSp}(4, 3)$ it is 81-dimensional and is $H_1(W(3, 3))$.

Stabiliser state A joint eigenstate of a maximal commuting set of Paulis.

Strongly regular graph A regular graph in which adjacent pairs have λ , and non-adjacent pairs μ , common neighbours.

Symplectic form An alternating, nondegenerate bilinear form; here the commutation phase of Pauli operators.

Tetracode The ternary $[4, 2, 3]$ code; here the evaluation code of the four clocks.

Transvection The symplectic map $x \mapsto x + \langle x, v \rangle v$.

Tripotent An element e of a Jordan triple system with $\{e, e, e\} = 2e$ (normalisation of this paper).

Witting polytope A regular complex polytope in $\mathbb{C}^4$ with 240 vertices, the E_8 roots, and 40 diameters.

References

- [1] J. F. Adams, *Lectures on Exceptional Lie Groups*, University of Chicago Press, 1996.
- [2] J. C. Baez, The octonions, *Bull. Amer. Math. Soc.* **39** (2002) 145–205.
- [3] J. C. Baez and J. Huerta, Division algebras and supersymmetry I, in *Superstrings, Geometry, Topology, and C^* -algebras*, Proc. Symp. Pure Math. **81** (2010) 65–80, arXiv:0909.0551.
- [4] L. Borsten, D. Dahanayake, M. J. Duff, H. Ebrahim and W. Rubens, Black holes, qubits and octonions, *Phys. Rep.* **471** (2009) 113–219, arXiv:0809.4685.
- [5] A. E. Brouwer and H. Van Maldeghem, *Strongly Regular Graphs*, Cambridge University Press, 2022.
- [6] A. Connes and C. Rovelli, Von Neumann algebra automorphisms and time–thermodynamics relation in generally covariant quantum theories, *Class. Quantum Grav.* **11** (1994) 2899–2918.
- [7] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1999.
- [8] L. Dixon, J. A. Harvey, C. Vafa and E. Witten, Strings on orbifolds, *Nucl. Phys.* **B261** (1985) 678–686.
- [9] M. J. Duff and S. Ferrara, E_7 and the tripartite entanglement of seven qubits, *Phys. Rev. D* **76** (2007) 025018, arXiv:quant-ph/0609227.
- [10] J. Faraut and A. Korányi, *Analysis on Symmetric Cones*, Oxford University Press, 1994.
- [11] H. Freudenthal, Beziehungen der E_7 und E_8 zur Oktavenebene I, *Indag. Math.* **16** (1954) 218–230.
- [12] S. Garibaldi, E_8 , the most exceptional group, *Bull. Amer. Math. Soc.* **53** (2016) 643–671.
- [13] K. S. Gibbons, M. J. Hoffman and W. K. Wootters, Discrete phase space based on finite fields, *Phys. Rev. A* **70** (2004) 062101.
- [14] M. Günaydin, K. Koepsell and H. Nicolai, Conformal and quasiconformal realizations of exceptional Lie groups, *Commun. Math. Phys.* **221** (2001) 57–76, arXiv:hep-th/0008063.
- [15] M. Günaydin, G. Sierra and P. K. Townsend, Exceptional supergravity theories and the magic square, *Phys. Lett.* **B133** (1983) 72–76.
- [16] M. Günaydin, Generalized conformal and superconformal group actions and Jordan algebras, *Mod. Phys. Lett. A* **8** (1993) 1407–1416, arXiv:hep-th/9301050.
- [17] M. Howard, J. Wallman, V. Veitch and J. Emerson, Contextuality supplies the ‘magic’ for quantum computation, *Nature* **510** (2014) 351–355.
- [18] J. E. Humphreys, The Steinberg representation, *Bull. Amer. Math. Soc.* **16** (1987) 247–263.
- [19] P. Imany, J. A. Jaramillo-Villegas, M. S. Alshaykh, J. M. Lukens, O. D. Odele, A. J. Moore, D. E. Leaird, M. Qi and A. M. Weiner, High-dimensional optical quantum logic in large operational spaces, *npj Quantum Inf.* **5** (2019) 59.
- [20] R. Kallosh and B. Kol, E_7 symmetric area of the black hole horizon, *Phys. Rev. D* **53** (1996) R5344, arXiv:hep-th/9602014.
- [21] H. Kraft, A. Regeta and S. Zimmermann, Small G -varieties, arXiv:2009.05559.

- [22] S. Krutelevich, Jordan algebras, exceptional groups and Bhargava composition, *J. Algebra* **314** (2007) 924–977.
- [23] T. Kugo and P. Townsend, Supersymmetry and the division algebras, *Nucl. Phys.* **B221** (1983) 357–380.
- [24] R. Landauer, Irreversibility and heat generation in the computing process, *IBM J. Res. Dev.* **5** (1961) 183–191.
- [25] O. Lebedev, H. P. Nilles, S. Raby, S. Ramos-Sánchez, M. Ratz, P. K. S. Vaudrevange and A. Wingerter, The heterotic road to the MSSM with R parity, *Phys. Rev. D* **77** (2008) 046013, arXiv:0708.2691.
- [26] S. Lloyd and S. L. Braunstein, Quantum computation over continuous variables, *Phys. Rev. Lett.* **82** (1999) 1784–1787.
- [27] C. A. Manogue, T. Dray and R. A. Wilson, Octonions: an E_8 description of the Standard Model, *J. Math. Phys.* **63** (2022) 081703, arXiv:2204.05310.
- [28] S. P. Martin, Some simple criteria for gauged R-parity, *Phys. Rev. D* **46** (1992) R2769, arXiv:hep-ph/9207218.
- [29] H. P. Nilles, S. Ramos-Sánchez, P. K. S. Vaudrevange and A. Wingerter, The Orbifolder: a tool to study the low-energy effective theory of heterotic orbifolds, *Comput. Phys. Commun.* **183** (2012) 1363–1380, arXiv:1110.5229.
- [30] D. N. Page and W. K. Wootters, Evolution without evolution: dynamics described by stationary observables, *Phys. Rev. D* **27** (1983) 2885–2892.
- [31] S. E. Payne and J. A. Thas, *Finite Generalized Quadrangles*, 2nd ed., European Mathematical Society, 2009.
- [32] T. A. Springer, Regular elements of finite reflection groups, *Invent. Math.* **25** (1974) 159–198.
- [33] M. Tegmark, On the dimensionality of spacetime, *Class. Quantum Grav.* **14** (1997) L69–L75, arXiv:gr-qc/9702052.
- [34] Watanabe, Belfiore, de Carvalho and Vieira Filho, a cyclotomic construction of E_8 from the negacyclic tetracode, *Int. J. Appl. Math.* **31**(1) (2018), doi:10.12732/ijam.v31i1.6 (as cited in Pass 10948).